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MIL-M-28787/359(SH)
9 March 1990

# MILITARY SPECIFICATION

## MODULES, DIGITAL, STANDARD ELECTRONIC

## FLIP/FLOP, D TYPE MODULE, KEY CODE KLR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic D type flip/flop module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

DOD-HDBK-217 - Reliability Predictions of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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## 3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.0662 failures per million operating hours when predicted in accordance with DOD-HDBK-217E.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be KLR.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned 1 through 40.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.



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TABLE I Module electrical requirements

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | 25 ± 5°C 4/ | | 0 ± 5°C 4/ | | 60 ± 5°C 4/ | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| A | Isolation, initial | II | S | 10 | INF | 10 | INF | 10 | INF | MOhms |
| | end-of-life | II | Q | 1 | INF | 1 | INF | 1 | INF | MOhms |
| B | Vcc stress | III | S | ---------Module shall pass all remaining tests--------- | | | | | | |
| C1-C6 | Ibd | IV | S | 0 | 100 | 0 | 100 | 0 | 100 | uA |
| C7-C18 | Ibd | IV | S | 0 | 200 | 0 | 200 | 0 | 200 | uA |
| C19-C24 | Ibd | IV | S | 0 | 300 | 0 | 300 | 0 | 300 | uA |
| D1-D12 | Ios | V | S | -100 | -15 | -100 | -15 | -100 | -15 | mA |
| E1-E24 | Vic | VI | S | -1.5 | -0.4 | -1 5 | -0.4 | -1.5 | -0.4 | V |
| F1-F24 | Vol | VII | X | 0.0 | 0.4 | 0 0 | 0.4 | 0.0 | 0.4 | V |
| G1-G36 | Voh | VII | X | 2.5 | 4.5 | 2 5 | 4.5 | 2.5 | 4.5 | V |
| G37,G38 | Voh | VII | X | 3.8 | 4.0 | 3 8 | 4.0 | 3.8 | 4.0 | V |
| H1-H6 | Iil | VIII | X | -0.40 | -0.03 | -0.40 | -0.03 | -0.40 | -0.03 | mA |
| H7-H18 | Iil | VIII | X | -0.80 | -0.06 | -0.80 | -0.06 | -0.80 | -0.06 | mA |
| H19-H24 | Iil | VIII | X | -1.20 | -0.06 | -1 20 | -0.06 | -1.20 | -0.06 | mA |
| J1-J6 | Iih | IX | X | 0 | 20 | 0 | 20 | 0 | 20 | uA |
| J7-J18 | Iih | IX | X | 0 | 40 | 0 | 40 | 0 | 40 | uA |
| J19-J24 | Iih | IX | X | 0 | 60 | 0 | 60 | 0 | 60 | uA |
| K1 | Icc | X | X | 1 | 36 | 1 | 36 | 1 | 36 | mA |
| K2 | Icc | X | X | 1 | 36 | 1 | 36 | 1 | 36 | mA |
| L1-L12 | Propagation delay time | XI | S | 5 | 57 | 5 | 70 | 5 | 70 | nS |
| M1-M12 | Propagation delay time | XII | S | 5 | 46 | 5 | 55 | 5 | 55 | nS |

- See footnotes at end of table





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TABLE I. Module electrical requirements - Continued.

| Test cond | Parameter 5/ | Ref test table | Test type 1/-3/ | 25 ± 5°C 4/ | | 0 ± 5°C 4/ | | 60 ± 5°C 4/ | | Units |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | |
| N1-N12 | Propagation delay time | XIII | S | 5 | 57 | 5 | 70 | 5 | 70 | nS |
| P1-P12 | Propagation delay time | XIV | S | 5 | 46 | 5 | 55 | 5 | 55 | nS |
| Q1-Q12 | Setup time, TA | XV | S | 900 | 1100 | 900 | 1100 | 900 | 1100 | nS |
| R1-R12 | Hold time, TA | XVI | S | 900 | 1100 | 900 | 1100 | 900 | 1100 | nS |
| S1-S12 | Maximum clock freq./ minimum clock pulse width, TA | XVII | S | 92 | 108 | 92 | 108 | 92 | 108 | nS |
| T1-T12 | Threshold clock, TA | XVIII | S | 900 | 1100 | 900 | 1100 | 900 | 1100 | nS |
| U1-U24 | Output Crosstalk, low and high level | XIX | Q | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| V | PSTA | XX | Q | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| W | PSFC | XXI | S | 3.6 | 6.0 | 3.6 | 6.0 | 3.6 | 6.0 | uF |
| X1-X24 | Cin | XXI | Q | 2 | 18 | 2 | 18 | 2 | 18 | pF |
| Y1-Y32 | Functional | XXII | X | ----------See table XXII---------- | | | | | | |

- See footnotes on next page.



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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.



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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/359. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram and pin assignments shall conform to figure 1.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

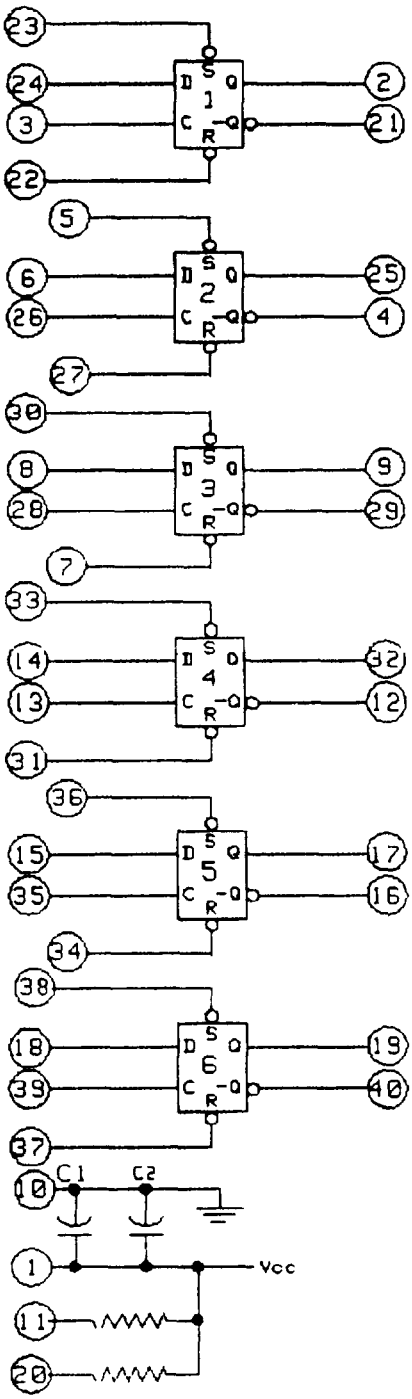
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.



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| | | | |
|---|---|---|---|
| -1Q | 21 | 1 | Vcc |
| 1R | 22 | 2 | 1Q |
| 1S | 23 | 3 | 1C |
| 1D | 24 | 4 | -2Q |
| 2Q | 25 | 5 | 2S |
| 2C | 26 | 6 | 2D |
| 2R | 27 | 7 | 3R |
| 3C | 28 | 8 | 3D |
| -3Q | 29 | 9 | 3Q |
| 3S | 30 | 10 | GND |
| 4R | 31 | 11 | PULL-UP RESISTOR |
| 4Q | 32 | 12 | -4Q |
| 4S | 33 | 13 | 4C |
| 5R | 34 | 14 | 4D |
| 5C | 35 | 15 | 5D |
| 5S | 36 | 16 | -5Q |
| 6R | 37 | 17 | 5Q |
| 6S | 38 | 18 | 6D |
| 6C | 39 | 19 | 6Q |
| -6Q | 40 | 20 | PULL-UP RESISTOR |

FIGURE 1. KLR functional block diagram and pin assignment.



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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. Test conditions A, B, C, D and E are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D and E where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

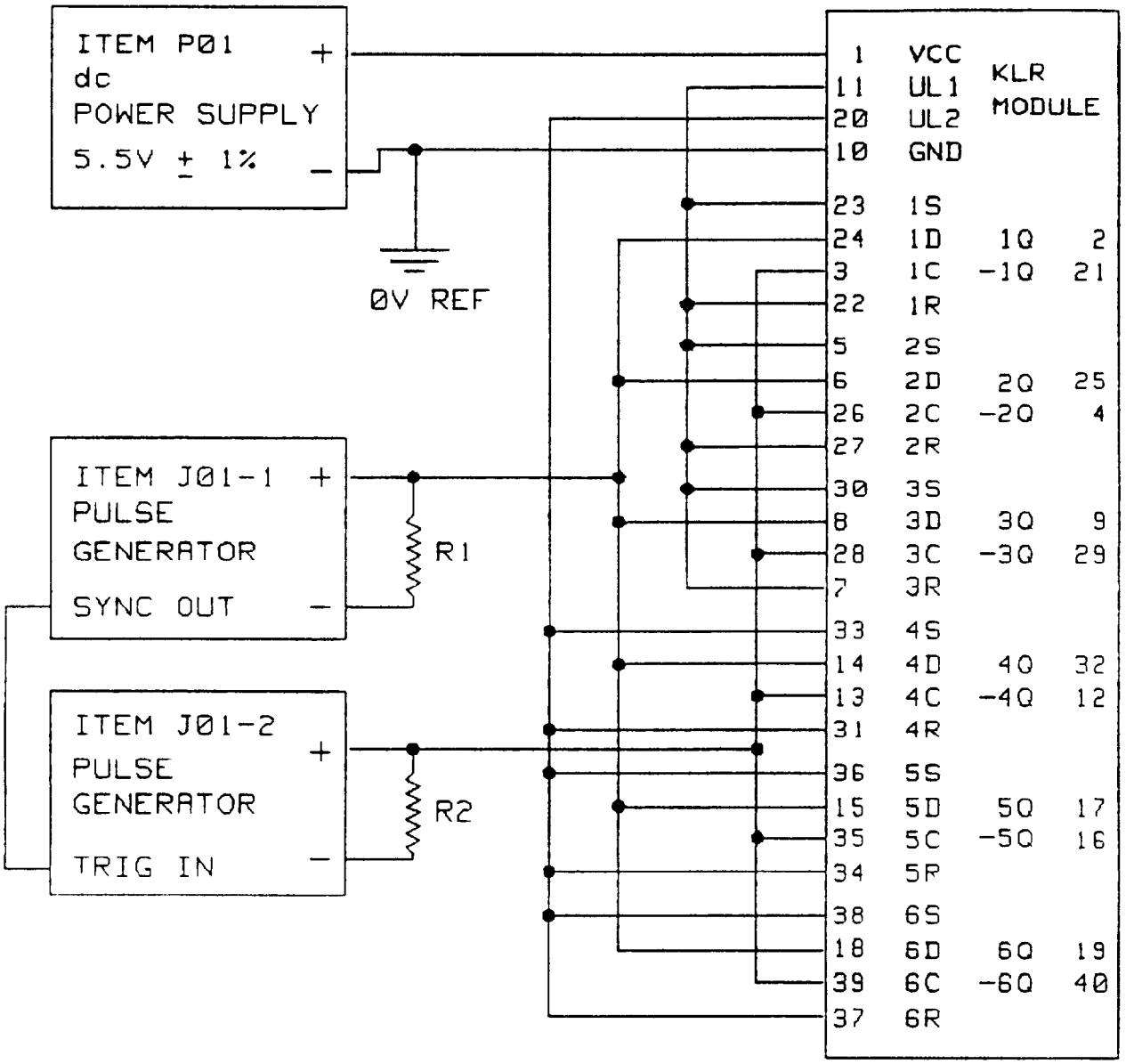
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions F through W and Y shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figures 2 and 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition W, shall be performed using the initial limits at 25 ± 5°C ambient.



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NOTES:

1. Load outputs as indicated in the output voltage test.
2. R1 = R2 = 51 ohms ± 5 percent, 1 W.

FIGURE 2. KLR operating temperature and life test circuit.



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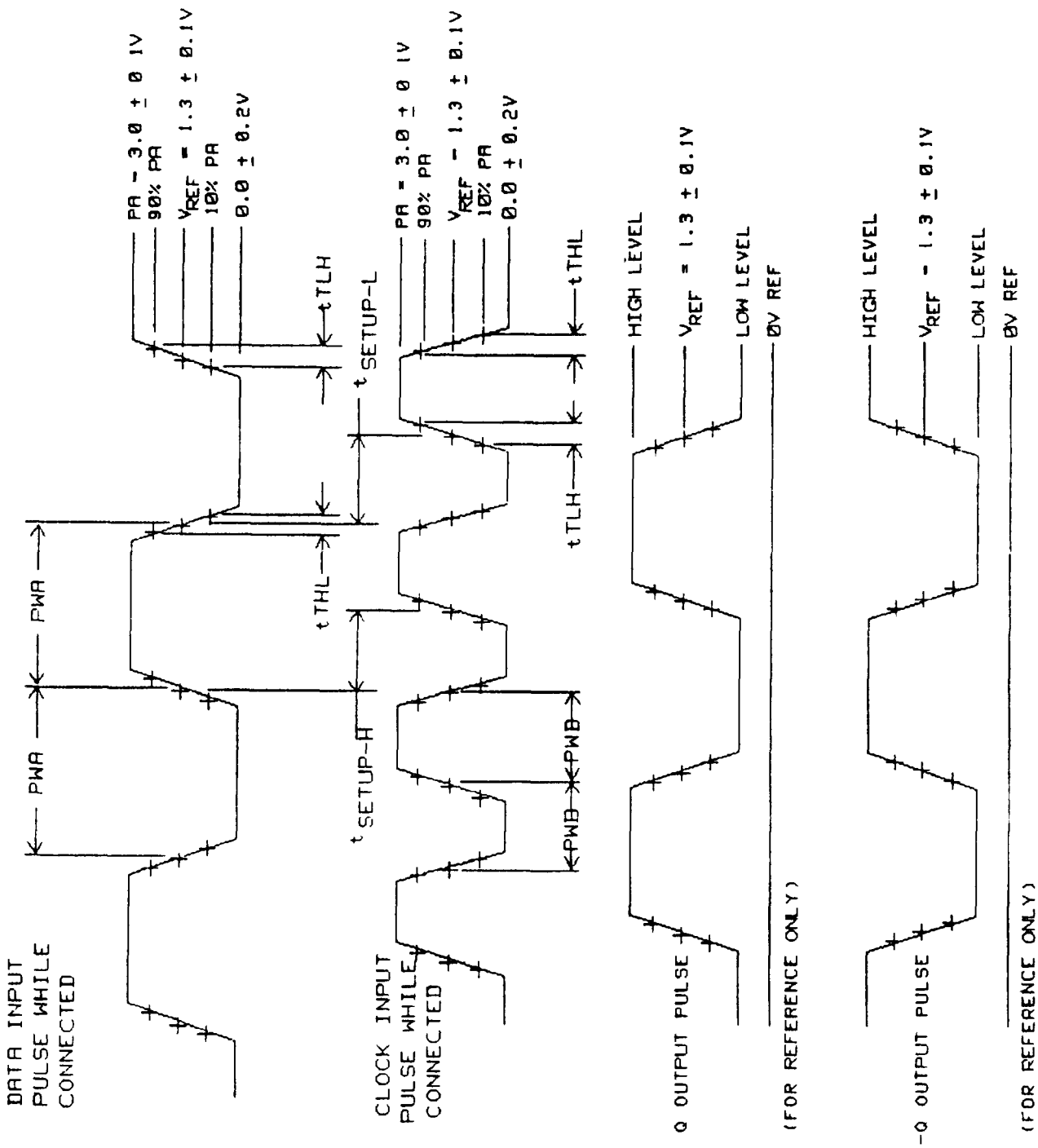



FIGURE 3. Operating temperature and life test waveforms.



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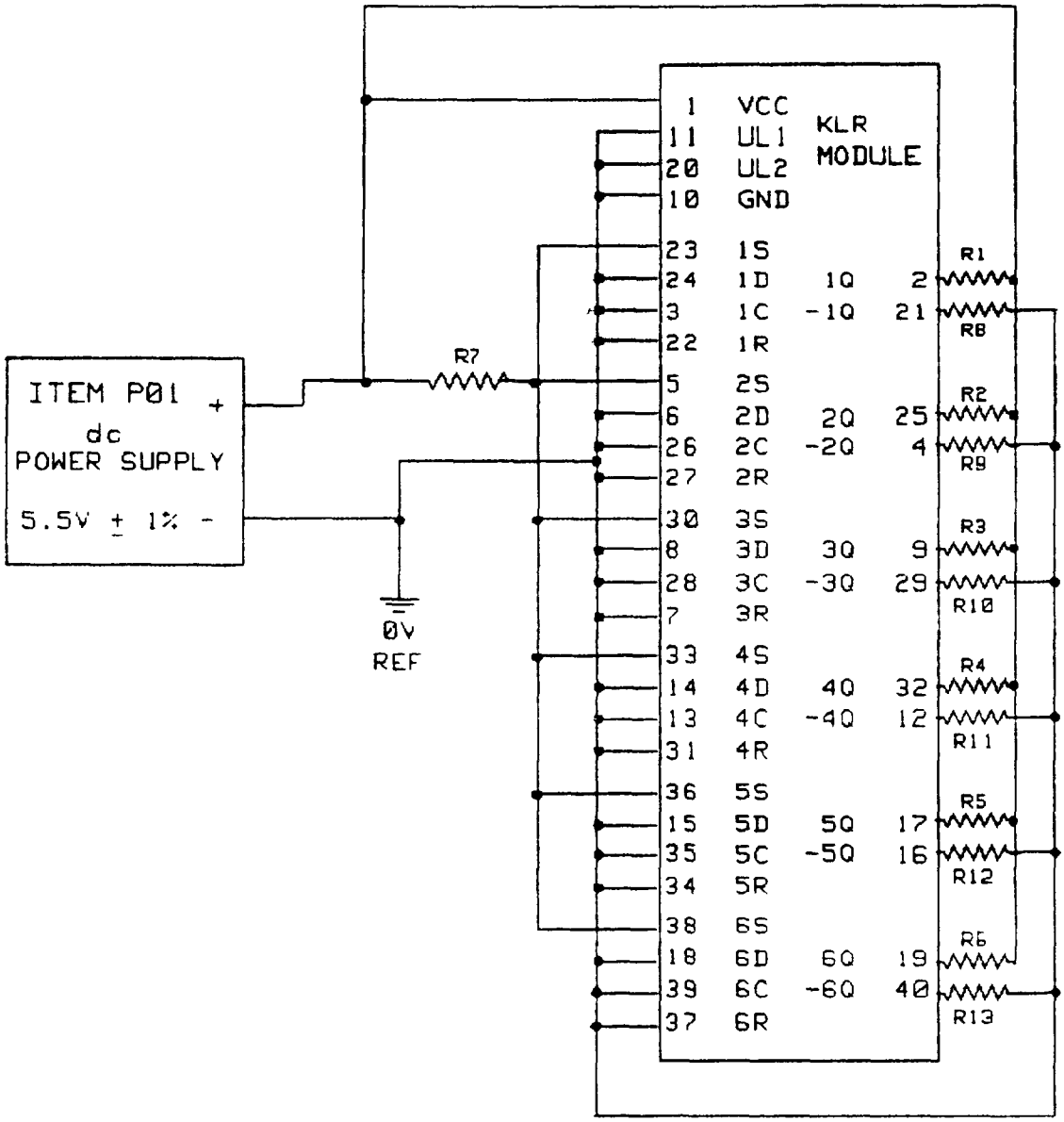
NOTES:

1. Input pulse parameters:

   PWA = 1000 ± 100 nS.
   PWB = 500 ± 50 nS.
   tTLH = tTHL = 10 ± 2 nS.
   tSETUP-H = tSETUP-L = 250 ± 25 nS.



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NOTES:

1. R1 = R2 = R3 = R4 = R5 = R6 = 1.27 Kohm ± 1 percent, 1/4 W.

2. R7 = 1 Kohm ± 1 percent, 1/4 W.

3. R8 = R9 = R10 = R11 = R12 = R13 = 6.19 Kohm ± 1 percent, 1/8 W.

FIGURE 4. KLR component temperature test circuit.



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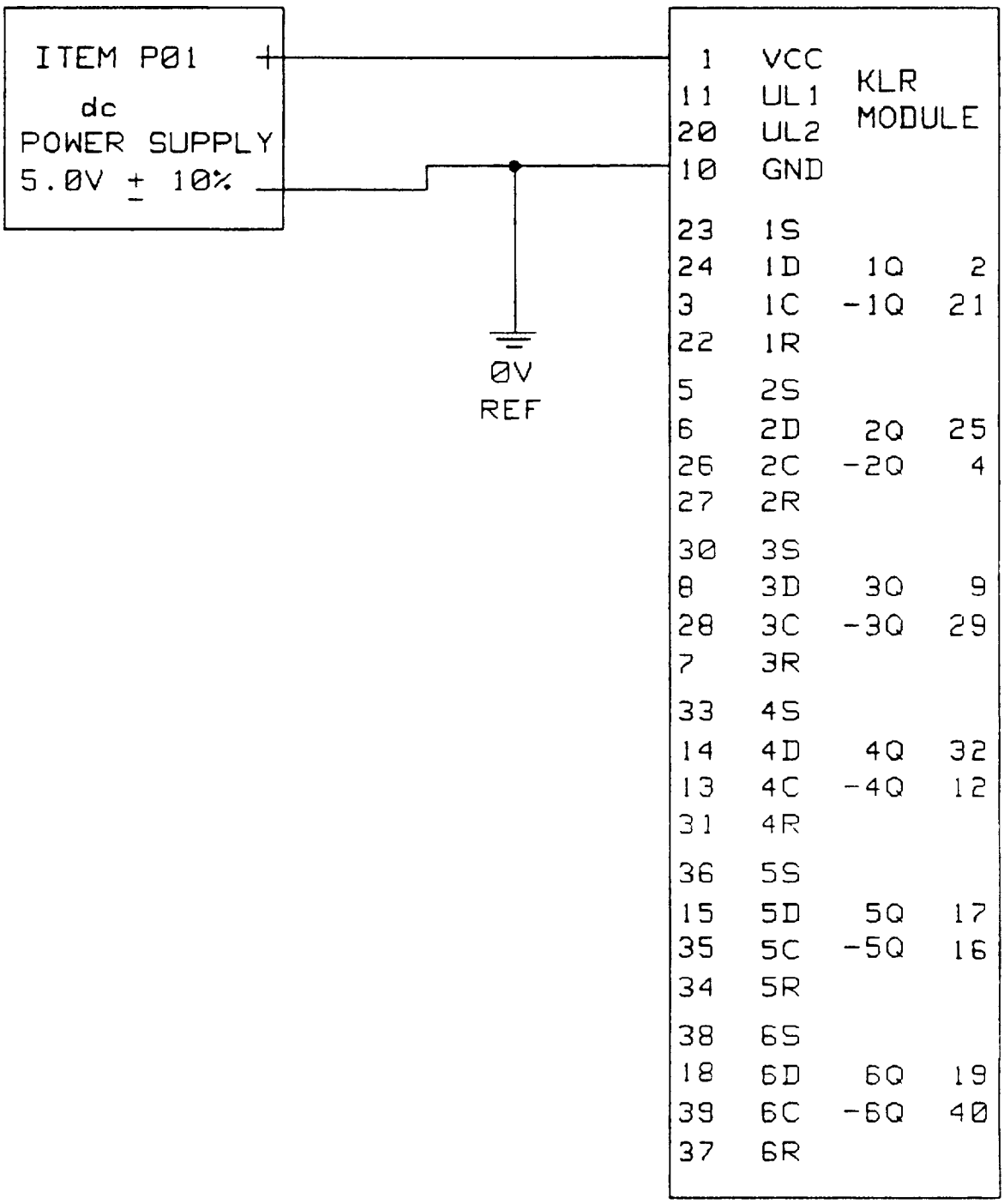



FIGURE 5. KLR vibration test circuit.



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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 2 and 3. Following the life test and while still at the maximum class temperature, electrical tests F through W and Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests F through W and Y shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ibd = input breakdown current.
(c) Icc = power supply current.
(d) Iih = input high level current.
(e) Iil = input low level current.
(f) Ioh = output high level current.
(g) Iol = output low level current.
(h) Ios = output short circuit current.
(i) NA = not applicable.
(j) PSFC = power supply filter capacitance.
(k) PSTA = power supply transient amplitude.
(l) Vic = input clamp diode voltage.
(m) Vih = input high level voltage.
(n) Vil = input low level voltage.
(o) Voh = output high level voltage.
(p) Vol = output low level voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 0.20 W (max).



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6.3.1 Input parameters.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil min (mA) | Iih max (uA) | Ibd max (uA) | Cin max (pF) |
|---|---|---|---|---|---|---|---|
| 1D | -1.5 | 0.7 | 2.0 | -0.40 | 20 | 100 | 18 |
| 2D | -1.5 | 0.7 | 2.0 | -0.40 | 20 | 100 | 18 |
| 3D | -1.5 | 0.7 | 2.0 | -0.40 | 20 | 100 | 18 |
| 4D | -1.5 | 0.7 | 2.0 | -0.40 | 20 | 100 | 18 |
| 5D | -1.5 | 0.7 | 2.0 | -0.40 | 20 | 100 | 18 |
| 6D | -1.5 | 0.7 | 2.0 | -0.40 | 20 | 100 | 18 |
| 1S | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 2S | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 3S | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 4S | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 5S | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 6S | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 1C | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 2C | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |



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6.3.1 Input parameters - Continued.

| Signal | Vic min (V) | Vil max (V) | Vih min (V) | Iil min (mA) | Iih max (uA) | Ibd max (uA) | Cin max (pF) |
|---|---|---|---|---|---|---|---|
| 3C | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 4C | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 5C | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 6C | -1.5 | 0.7 | 2.0 | -0.80 | 40 | 200 | 18 |
| 1R | -1.5 | 0.7 | 2.0 | -1.20 | 60 | 300 | 18 |
| 2R | -1.5 | 0.7 | 2.0 | -1.20 | 60 | 300 | 18 |
| 3R | -1.5 | 0.7 | 2.0 | -1.20 | 60 | 300 | 18 |
| 4R | -1.5 | 0.7 | 2.0 | -1.20 | 60 | 300 | 18 |
| 5R | -1.5 | 0.7 | 2.0 | -1.20 | 60 | 300 | 18 |
| 6R | -1.5 | 0.7 | 2.0 | -1.20 | 60 | 300 | 18 |



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6.3.2 Output parameters.

| Signal | Voh min (V) | Vol max (V) | Ioh min (uA) | Iol max (mA) | Ios max (mA) |
|---|---|---|---|---|---|
| 1Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| 2Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| 3Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| 4Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| 5Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| 6Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| -1Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| -2Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| -3Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| -4Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| -5Q | 2.5 | 0.4 | -400 | 4.0 | -15 |
| -6Q | 2.5 | 0.4 | -400 | 4.0 | -15 |

6.4 Module function. The module function is printed herein in the appendix.



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6.4.1 Description of I/O pins.

| Signal name | Pin number | I/O | Functional description |
|---|---|---|---|
| 1D | 24 | I | Clocked data inputs. |
| 2D | 6 | I | |
| 3D | 8 | I | |
| 4D | 14 | I | |
| 5D | 15 | I | |
| 6D | 18 | I | |
| 1C | 3 | I | Clock inputs. |
| 2C | 26 | I | |
| 3C | 28 | I | |
| 4C | 13 | I | |
| 5C | 35 | I | |
| 6C | 39 | I | |
| 1S | 23 | I | Flip/flop set inputs. |
| 2S | 5 | I | |
| 3S | 30 | I | |
| 4S | 33 | I | |
| 5S | 36 | I | |
| 6S | 38 | I | |
| 1R | 22 | I | Flip/flop reset inputs. |
| 2R | 27 | I | |
| 3R | 7 | I | |
| 4R | 31 | I | |
| 5R | 34 | I | |
| 6R | 37 | I | |
| 1Q | 2 | O | Flip/flop outputs and their respective inversions. |
| -1Q | 21 | O | |
| 2Q | 25 | O | |
| -2Q | 4 | O | |
| 3Q | 9 | O | |
| -3Q | 29 | O | |
| 4Q | 32 | O | |
| -4Q | 12 | O | |
| 5Q | 17 | O | |
| -5Q | 16 | O | |
| 6Q | 19 | O | |
| -6Q | 40 | O | |



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6.4.1 Description of I/O pins - Continued.

| Signal name | Pin number | I/O | Functional description |
|---|---|---|---|
| UL1<br>UL2 | 11<br>20 | I/O<br>I/O | Pull up resistors. |

6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II, figure 6.

(b) Vcc stress test - table III, figure 7.

(c) Ibd test - table IV, figure 8.

(d) Ios test - table V, figure 9.

(e) Vic test - table VI, figure 10.

(f) Vol and Voh tests - table VII, figures 11 and 12.

(g) Iil test - table VIII, figure 13.

(h) Iih test - table IX, figure 13.

(i) Icc test - table X, figure 14.

(j) Propagation delay time tests - tables XI through XIV, figures 15 through 18.

(k) Setup time test - table XV, figures 19 and 20.

(l) Hold time test - table XVI, figures 19 and 21.

(m) Clock frequency test - table XVII, figures 22 and 23.

(n) Clock threshold test - table XVIII, figures 22 and 24.

(o) Output crosstalk test - table XIX, figures 25 and 26.



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(p) PSTA test - table XX, figure 27 and 28.

(q) PSFC test - table XXI, figure 29.

(r) Cin test - table XXI, figure 29.

(s) Functional test - table XXII, figure 30.

Preparing activity:
Navy - SH

(Project 5963-N744)

Agent:
NWSC Crane



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TABLE II. Isolation test.

| Test cond | Test description | Connect A to pin | 25 ± 5°C Initial limits Min | Max | Units | Notes | Fig |
|---|---|---|---|---|---|---|---|
| A | Isolation, frame | Frame | 10 | INF | MOhms | 1-2 | 6 |

NOTES:

1. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

2. Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.



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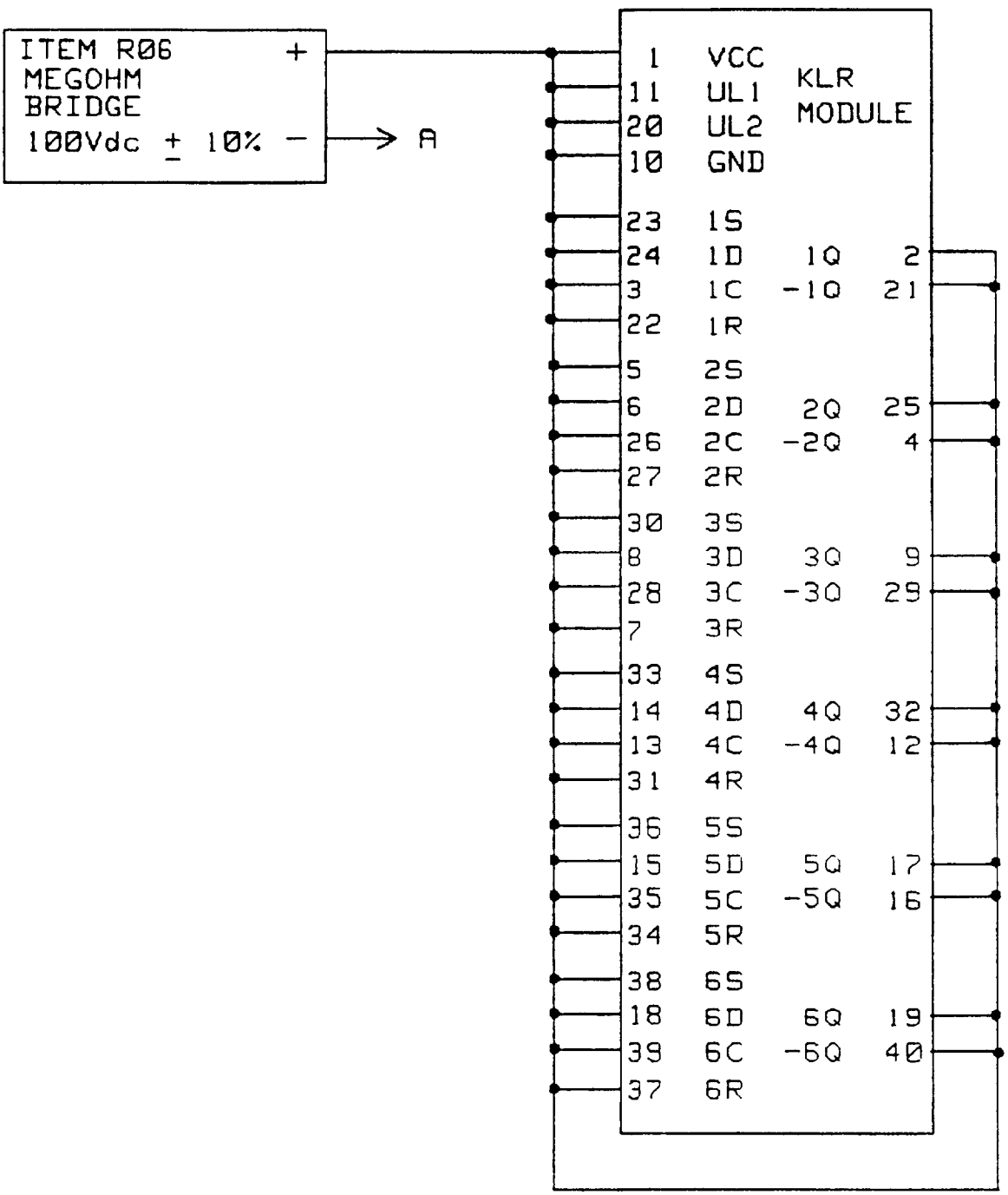



FIGURE 6. Isolation test circuit for table II.



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TABLE III. Power supply stress test.

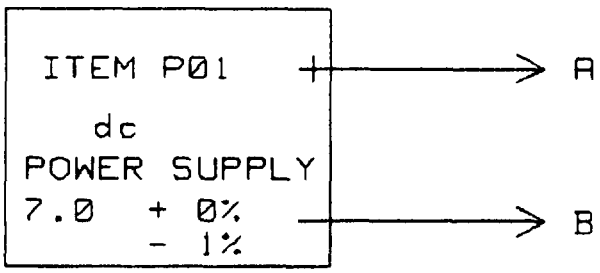
| Test cond | Power supply | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits | Notes | Fig |
|---|---|---|---|---|---|---|
| B | Vcc | 1 | 10 | Module shall pass all remaining tests. | 1 | 7 |

NOTES:

1. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.



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| | | | |
|---|---|---|---|
| 1 | VCC | KLR MODULE | |
| 11 | UL1 | | |
| 20 | UL2 | | |
| 10 | GND | | |
| 23 | 1S | | |
| 24 | 1D | 1Q | 2 |
| 3 | 1C | -1Q | 21 |
| 22 | 1R | | |
| 5 | 2S | | |
| 6 | 2D | 2Q | 25 |
| 26 | 2C | -2Q | 4 |
| 27 | 2R | | |
| 30 | 3S | | |
| 8 | 3D | 3Q | 9 |
| 28 | 3C | -3Q | 29 |
| 7 | 3R | | |
| 33 | 4S | | |
| 14 | 4D | 4Q | 32 |
| 13 | 4C | -4Q | 12 |
| 31 | 4R | | |
| 36 | 5S | | |
| 15 | 5D | 5Q | 17 |
| 35 | 5C | -5Q | 16 |
| 34 | 5R | | |
| 38 | 6S | | |
| 18 | 6D | 6Q | 19 |
| 39 | 6C | -6Q | 40 |
| 37 | 6R | | |

FIGURE 7. Vcc stress test circuit for table III.



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TABLE IV. Input breakdown current test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Fig |
|---|---|---|---|---|---|---|---|---|---|---|
| C1 | 1D | 24 | 22 | Open | 3,23 | 0 | 100 | uA | 1-3 | 8 |
| C2 | 2D | 6 | 27 | | 5,26 | | | | | |
| C3 | 3D | 8 | 7 | | 28,30 | | | | | |
| C4 | 4D | 14 | 31 | | 13,33 | | | | | |
| C5 | 5D | 15 | 34 | | 35,36 | | | | | |
| C6 | 6D | 18 | 37 | | 38,39 | 0 | 100 | uA | 1-3 | 8 |
| C7 | 1C | 3 | 22 | | 23,24 | 0 | 200 | uA | 1-3 | 8 |
| C8 | 2C | 26 | 27 | | 5,6 | | | | | |
| C9 | 3C | 28 | 7 | | 8,30 | | | | | |
| C10 | 4C | 13 | 31 | | 14,33 | | | | | |
| C11 | 5C | 35 | 34 | | 15,36 | | | | | |
| C12 | 6C | 39 | 37 | Open | 18,38 | | | | | |
| C13 | 1S | 23 | Open | 3 | 22,24 | | | | | |
| C14 | 2S | 5 | | 26 | 6,27 | | | | | |
| C15 | 3S | 30 | | 28 | 7,8 | | | | | |
| C16 | 4S | 33 | | 13 | 14,31 | | | | | |
| C17 | 5S | 36 | | 35 | 15,34 | | | | | |
| C18 | 6S | 38 | Open | 39 | 18,37 | 0 | 200 | uA | 1-3 | 8 |
| C19 | 1R | 22 | 24 | 3 | 23 | 0 | 300 | uA | 1-3 | 8 |
| C20 | 2R | 27 | 6 | 26 | 5 | 0 | 300 | uA | 1-3 | 8 |



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TABLE IV. Input breakdown current test - Continued.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Fig |
|---|---|---|---|---|---|---|---|---|---|---|
| C21 | 3R | 7 | 8 | 28 | 30 | 0 | 300 | uA | 1-3 | 8 |
| C22 | 4R | 31 | 14 | 13 | 33 | | | | | |
| C23 | 5R | 34 | 15 | 35 | 36 | | | | | |
| C24 | 6R | 37 | 18 | 39 | 38 | 0 | 300 | uA | 1-3 | 8 |

NOTES:

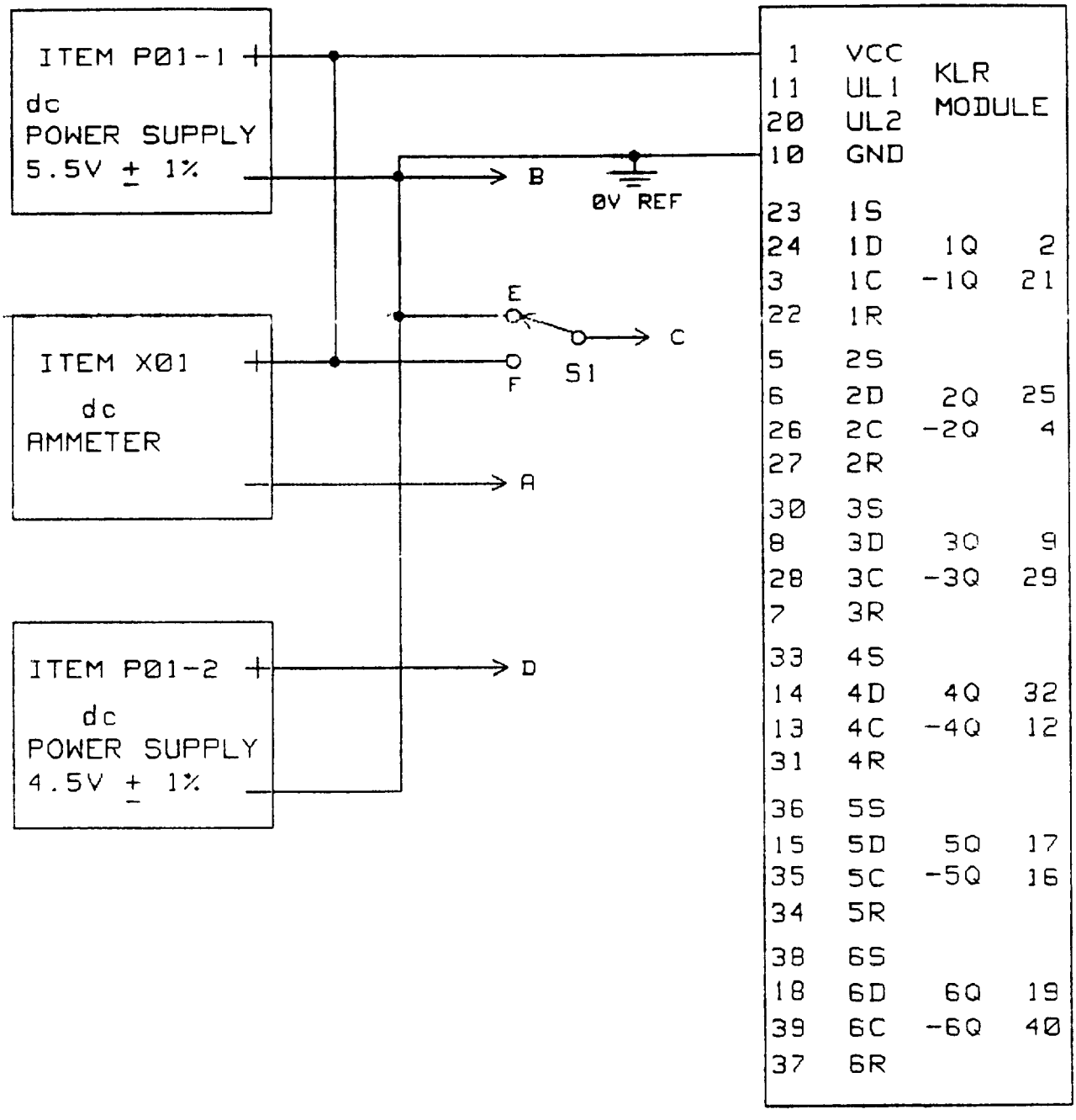
1. All input pins not under test, or conditioned for that test shall be connected to ground.

2. Caution: Do not isolate the chassis of ammeter X01 from earth ground.

3. For electrical requirements C13-C24, prior to making the measurement, set S1 in the F position, move S1 to the E position for a minimum of 100 nS, and return S1 to the F position, make the required measurement.



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NOTE:

1. S1 = DPST switch.

FIGURE 8. Input breakdown current test circuit for table IV.



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TABLE V. Output short circuit current test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Fig |
|---|---|---|---|---|---|---|---|---|---|
| D1 | 1Q | 3,5,6,7,8,13, 14,15,18,23, 24,26,27,28, 30,31,33,34, 35,36,37,38,39 | 2 | 22 | -100 | -15 | mA | 1-3 | 9 |
| D2 | -1Q | 3,5,6,7,8,13, 14,15,18,22, 24,26,27,28, 30,31,33,34, 35,36,37,38,39 | 21 | 23 | | | | | |
| D3 | 2Q | 3,5,6,7,8,13, 14,15,18,22, 23,24,26,28, 30,31,33,34, 35,36,37,38,39 | 25 | 27 | | | | | |
| D4 | -2Q | 3,6,7,8,13,14, 15,18,22,23, 24,26,27,28, 30,31,33,34, 35,36,37,38,39 | 4 | 5 | | | | | |
| D5 | 3Q | 3,5,6,8,13,14, 15,18,22,23, 24,26,27,28, 30,31,33,34, 35,36,37,38,39 | 9 | 7 | | | | | |
| D6 | -3Q | 3,5,6,7,8,13, 14,15,18,22, 23,24,26,27, 28,31,33,34, 35,36,37,38,39 | 29 | 30 | -100 | -15 | mA | 1-3 | 9 |



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TABLE V. Output short circuit current test - Continued.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Fig |
|---|---|---|---|---|---|---|---|---|---|
| D7 | 4Q | 3,5,6,7,8,13, 14,15,18,22, 23,24,26,27, 28,30,33,34, 35,36,37,38,39 | 32 | 31 | -100 | -15 | mA | 1-3 | 9 |
| D8 | -4Q | 3,5,6,7,8,13, 14,15,18,22, 23,24,26,27, 28,30,31,34, 35,36,37,38,39 | 12 | 33 | | | | | |
| D9 | 5Q | 3,5,6,7,8,13, 14,15,18,22, 23,24,26,27, 28,30,31,33, 35,36,37,38,39 | 17 | 34 | | | | | |
| D10 | -5Q | 3,5,6,7,8,13, 14,15,18,22, 23,24,26,27, 28,30,31,33, 34,35,37,38,39 | 16 | 36 | | | | | |
| D11 | 6Q | 3,5,6,7,8,13, 14,15,18,22, 23,24,26,27, 28,30,31,33, 34,35,36,38,39 | 19 | 37 | | | | | |
| D12 | -6Q | 3,5,6,7,8,13, 14,15,18,22, 23,24,26,27, 28,30,31,33, 34,35,36,37,39 | 40 | 38 | -100 | -15 | mA | 1-3 | 9 |



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NOTES:

1. Do not isolate the chassis of ammeter X01 from earth ground.

2. If tests are successively repeated, a time of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

3. When performing output short circuit tests, only one output shall be shorted at a time and the duration of the test shall not exceed 1 second.



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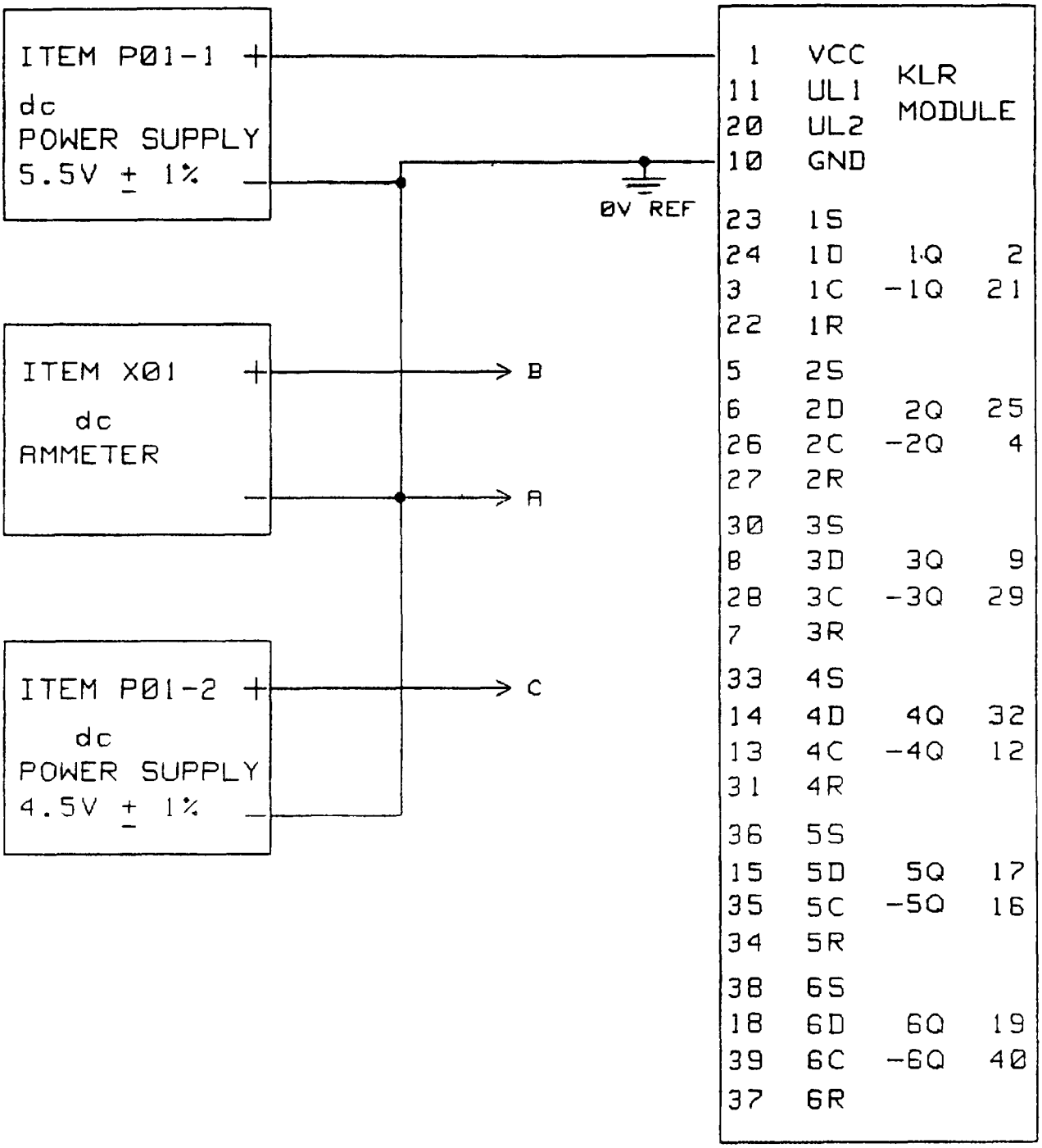



FIGURE 9. Output short circuit current test circuit for table V.



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TABLE VI. Input clamp diode voltage test.

| Test cond | Signal name | Connect A to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Fig |
|---|---|---|---|---|---|---|
| E1 | 1D | 24 | -1.5 | -0.4 | V | 10 |
| E2 | 2D | 6 | | | | |
| E3 | 3D | 8 | | | | |
| E4 | 4D | 14 | | | | |
| E5 | 5D | 15 | | | | |
| E6 | 6D | 18 | | | | |
| E7 | 1C | 3 | | | | |
| E8 | 2C | 26 | | | | |
| E9 | 3C | 28 | | | | |
| E10 | 4C | 13 | | | | |
| E11 | 5C | 35 | | | | |
| E12 | 6C | 39 | | | | |
| E13 | 1S | 23 | | | | |
| E14 | 2S | 5 | | | | |
| E15 | 3S | 30 | | | | |
| E16 | 4S | 33 | | | | |
| E17 | 5S | 36 | | | | |
| E18 | 6S | 38 | | | | |
| E19 | 1R | 22 | | | | |
| E20 | 2R | 27 | -1.5 | -0.4 | V | 10 |



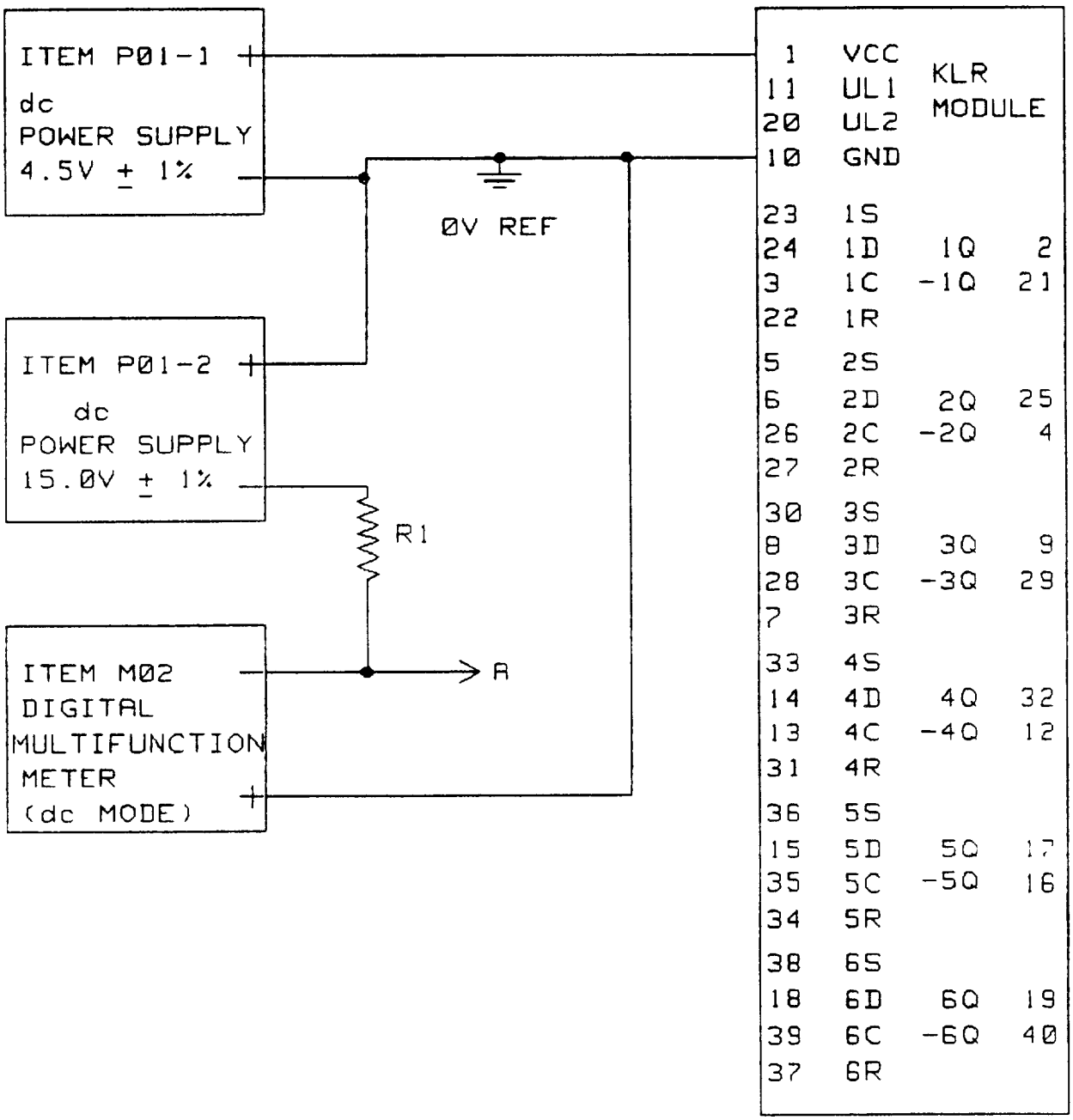
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TABLE VI. Input clamp diode voltage test - Continued.

| Test cond | Signal name | Connect A to pin | 25 ± 5°C Initial limits | | Units | Fig |
|---|---|---|---|---|---|---|
| | | | Min | Max | | |
| E21 | 3R | 7 | -1.5 | -0.4 | V | 10 |
| E22 | 4R | 31 | | | | |
| E23 | 5R | 34 | | | | |
| E24 | 6R | 37 | -1.5 | -0.4 | V | 10 |



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NOTE:

1. R1 = 750 Ohm ± 1 percent, 1/4 W.

FIGURE 10. Input clamp diode voltage test circuit for table VI.



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TABLE VII. Output voltage, high and low level, test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 + 5°C Initial limits Min | 25 + 5°C Initial limits Max | Units | Figs |
|---|---|---|---|---|---|---|---|---|---|
| G1 | -1Q (Voh) | 24,23 | Open | 22,3 | 21 | 2.5 | 4.5 | V | 11,12 |
| F1 | 1Q (Vol) | 24,23 | | 22,3 | 2 | 0.0 | 0.4 | V | 11,12 |
| G2 | 1Q (Voh) | 24,22 | | 23,3 | 2 | 2.5 | 4.5 | V | 11,12 |
| F2 | -1Q (Vol) | 24,22 | Open | 23,3 | 21 | 0.0 | 0.4 | V | 11,12 |
| G3 | 1Q (Voh) | 24,23, 22 | 3 | Open | 2 | 2.5 | 4.5 | V | 11,12 |
| F3 | -1Q (Vol) | 24,23, 22 | | Open | 21 | 0.0 | 0.4 | V | 11,12 |
| G4 | -1Q (Voh) | 23,22 | | 24 | 21 | 2.5 | 4.5 | V | 11,12 |
| F4 | 1Q (Vol) | 23,22 | 3 | 24 | 2 | 0.0 | 0.4 | V | 11,12 |
| G5 | 1Q (Voh) | Open | Open | 3,22, 23,24 | 2 | 2 5 | 4.5 | V | 11,12 |
| G6 | -1Q (Voh) | Open | | 3,22, 23,24 | 21 | 2.5 | 4.5 | V | 11,12 |
| G7 | -2Q (Voh) | 6,5 | | 27,26 | 4 | 2.5 | 4.5 | V | 11,12 |
| F5 | 2Q (Vol) | 6,5 | | 27,26 | 25 | 0.0 | 0.4 | V | 11,12 |
| G8 | 2Q (Voh) | 6,27 | | 5,26 | 25 | 2.5 | 4.5 | V | 11,12 |
| F6 | -2Q (Vol) | 6,27 | Open | 5,26 | 4 | 0.0 | 0.4 | V | 11,12 |
| G9 | 2Q (Voh) | 6,5,27 | 26 | Open | 25 | 2.5 | 4.5 | V | 11,12 |





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TABLE VII. Output voltage, high and low level, test - Continued.

| Test cond | Signal name | Connect | | | | 25 + 5°C Initial T limits | | Units | Figs |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| F7 | -2Q (Vol) | 6,5,27 | 26 | Open | 4 | 0 0 | 0.4 | V | 11,12 |
| G10 | -2Q (Voh) | 5,27 | | 6 | 4 | 2.5 | 4.5 | V | 11,12 |
| F8 | 2Q (Vol) | 5,27 | 26 | 6 | 25 | 0.0 | 0.4 | V | 11,12 |
| G11 | 2Q (Voh) | Open | Open | 5,6, 26,27 | 25 | 2 5 | 4.5 | V | 11,12 |
| G12 | -2Q (Voh) | Open | | 5,6, 26,27 | 4 | 2 5 | 4.5 | V | 11,12 |
| G13 | -3Q (Voh) | 8,30 | | 7,28 | 29 | 2.5 | 4.5 | V | 11,12 |
| F9 | 3Q (Vol) | 8,30 | Open | 7,28 | 9 | 0 0 | 0.4 | V | 11,12 |
| G14 | 3Q (Voh) | 8,7 | Open | 30,28 | 9 | 2.5 | 4.5 | V | 11,12 |
| F10 | -3Q (Vol) | 8,7 | Open | 30,28 | 29 | 0.0 | 0.4 | V | 11,12 |
| G15 | 3Q (Voh) | 8,30,7 | 28 | Open | 9 | 2.5 | 4.5 | V | 11,12 |
| F11 | -3Q (Vol) | 8,30,7 | | Open | 29 | 0.0 | 0.4 | V | 11,12 |
| G16 | -3Q (Voh) | 30,7 | | 8 | 29 | 2.5 | 4.5 | V | 11,12 |
| F12 | 3Q (Vol) | 30,7 | 28 | 8 | 9 | 0.0 | 0.4 | V | 11,12 |
| G17 | 3Q (Voh) | Open | Open | 7,8, 28,30 | 9 | 2.5 | 4.5 | V | 11,12 |





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TABLE VII. Output voltage, high and low level, test - Continued.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 + 5°C Initial limits Min | 25 + 5°C Initial limits Max | Units | Figs |
|---|---|---|---|---|---|---|---|---|---|
| G18 | -3Q (Voh) | Open | Open | 7,8, 28,30 | 29 | 2 5 | 4.5 | V | 11,12 |
| G19 | -4Q (Voh) | 14,33 | | 31,13 | 12 | 2.5 | 4.5 | V | 11,12 |
| F13 | 4Q (Vol) | 14,33 | | 31,13 | 32 | 0.0 | 0.4 | V | 11,12 |
| G20 | 4Q (Voh) | 14,31 | | 33,13 | 32 | 2.5 | 4.5 | V | 11,12 |
| F14 | -4Q (Vol) | 14,31 | Open | 33,13 | 12 | 0.0 | 0.4 | V | 11,12 |
| G21 | 4Q (Voh) | 14,33, 31 | 13 | Open | 32 | 2.5 | 4.5 | V | 11,12 |
| F15 | -4Q (Vol) | 14,33, 31 | | Open | 12 | 0.0 | 0.4 | V | 11,12 |
| G22 | -4Q (Voh) | 33,31 | | 14 | 12 | 2.5 | 4 5 | V | 11,12 |
| F16 | 4Q (Vol) | 33,31 | 13 | 14 | 32 | 0.0 | 0.4 | V | 11,12 |
| G23 | 4Q (Voh) | Open | Open | 13,14, 31,33 | 32 | 2.5 | 4 5 | V | 11,12 |
| G24 | -4Q (Voh) | Open | | 13,14 31,33 | 12 | 2.5 | 4.5 | V | 11,12 |
| G25 | -5Q (Voh) | 15,36 | | 34,35 | 16 | 2.5 | 4.5 | V | 11,12 |
| F17 | 5Q (Vol) | 15,36 | | 34,35 | 17 | 0.0 | 0.4 | V | 11,12 |
| G26 | 5Q (Voh) | 15,34 | | 36,35 | 17 | 2.5 | 4.5 | V | 11,12 |
| F18 | -5Q (Vol) | 15,34 | Open | 36,35 | 16 | 0.0 | 0.4 | V | 11,12 |





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TABLE VII Output voltage, high and ow level, test - Continued.

| Test cond | Signal name | Connect | | | | 25 ± 5°C Initial limits | | Units | Figs |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| G27 | 5Q (Voh) | 15,36, 34 | 35 | Open | 17 | 2 5 | 4.5 | V | 11, 12 |
| F19 | -5Q (Vol) | 15,36, 34 | 35 | Open | 16 | 0.0 | 0.4 | V | 11, 12 |
| G28 | -5Q (Voh) | 36,34 | 35 | 15 | 16 | 2.5 | 4 5 | V | 11, 12 |
| F20 | 5Q (Vol) | 36,34 | 35 | 15 | 17 | 0.0 | 0.4 | V | 11, 12 |
| G29 | 5Q (Voh) | Open | Open | 15,34, 35,36 | 17 | 2.5 | 4.5 | V | 11, 12 |
| G30 | -5Q (Voh) | Open | | 15,34 35,36 | 16 | 2.5 | 4.5 | V | 11, 12 |
| G31 | -6Q (Voh) | 18,38 | | 37,39 | 40 | 2.5 | 4.5 | V | 11, 12 |
| F21 | 5Q (Vol) | 18,38 | | 37,39 | 19 | 0.0 | 0.4 | V | 11, 12 |
| G32 | 6Q (Voh) | 18,37 | | 38,39 | 19 | 2 5 | 4.5 | V | 11, 12 |
| F22 | -6Q (Vol) | 18,37 | Open | 38,39 | 40 | 0 0 | 0 4 | V | 11, 12 |
| G33 | 6Q (Voh) | 18,38, 37 | 39 | Open | 19 | 2 5 | 4.5 | V | 11, 12 |
| F23 | -6Q (Vol) | 18,38, 37 | | Open | 40 | 0 0 | 0.4 | V | 11, 12 |
| G34 | -6Q (Voh) | 38,37 | | 18 | 40 | 2 5 | 4.5 | V | 11, 12 |
| F24 | 6Q (Vol) | 38,37 | 39 | 18 | 19 | 0 0 | 0 4 | V | 11, 12 |





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TABLE VII Output voltage, high and low level, test - Continued.

| Test cond | Signal name | Connect | | | | 25 ± 5°C Initial limits | | Units | Figs |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| G35 | 6Q (Voh) | Open | Open | 18,37, 38,39 | 19 | 2.5 | 4.5 | V | 11,12 |
| G36 | -6Q (Voh) | | | 18,37, 38,39 | 40 | 2.5 | 4.5 | V | 11,12 |
| G37 | UL1 (Voh) | | | Open | 11 | 3 8 | 4.0 | V | 11,12 |
| G38 | UL2 (Voh) | Open | Open | Open | 20 | 3 8 | 4.0 | V | 11,12 |



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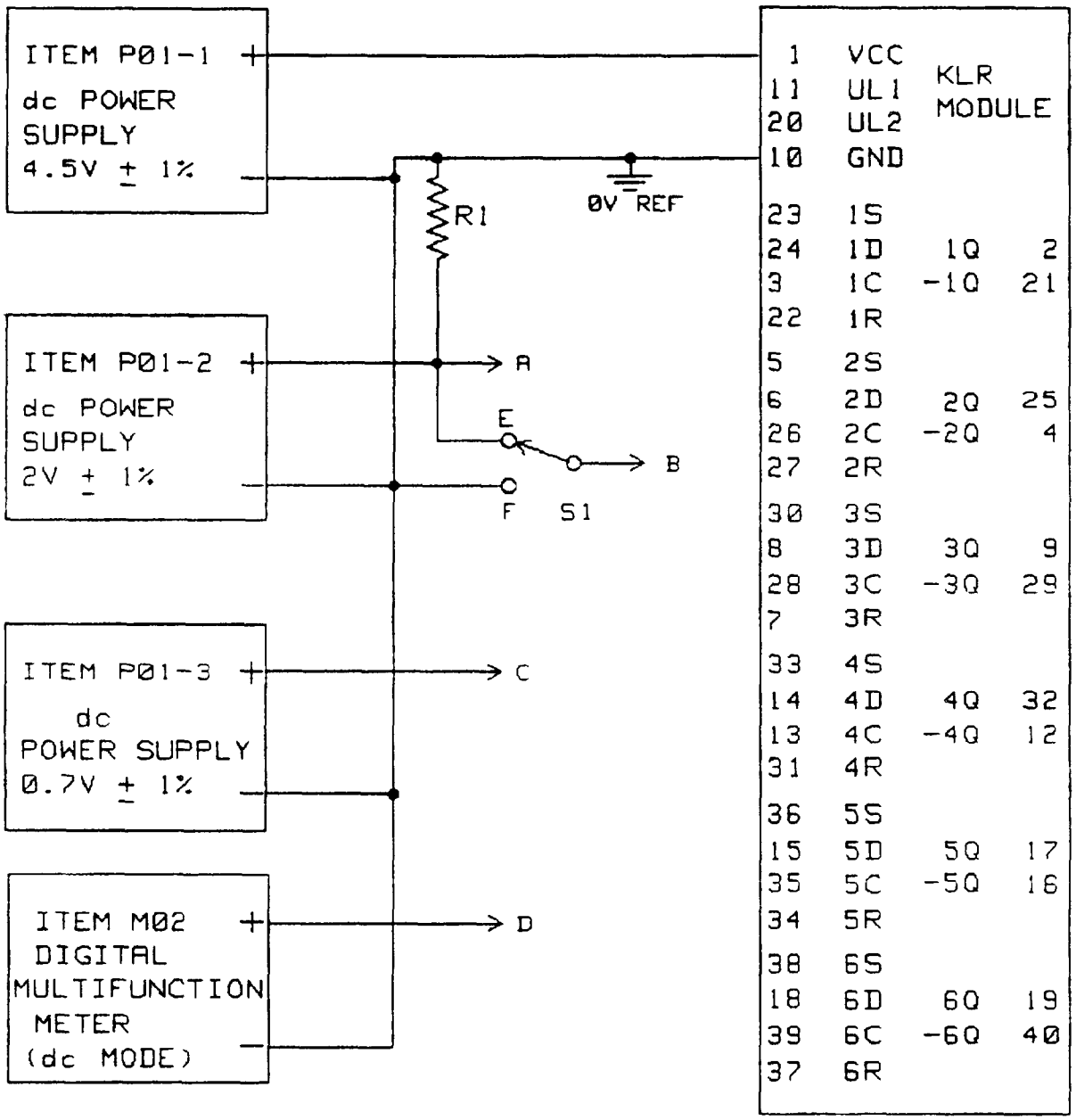
NOTES:

1. For test requirements that connect point B to a module pin, prior to making each measurement, set S1 in the E position; move S1 to the F position for a minimum of 100 nS; and return S1 to the E position and make the required measurement.

2. Load outputs as shown on figure 12.



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NOTES:

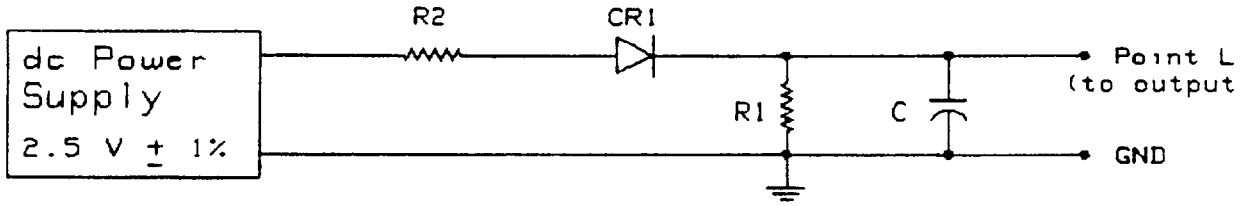
1. R1 = 51 Ohm ± 5 percent, 1/4 W.

2. S1 = DPST switch.

FIGURE 11. Output voltage, low-level and high-level, test circuit for table VII.



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NOTES:

1. CR1 = G321 (small case), 1N4150 or equivalent.

2. R1 = 6.19 KOhms ± 1 percent, 1/8 W.

3. R2 = select value for current flow of 4.0 mA ± 5 percent out of point L with point L held at 0.4 V.

4. C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk tests.

5. C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

6. C is not required when performing dc electrical tests.

7. The maximum lead length from point L to the appropriate module pin shall be 1 inch (25.4 mm).

8. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

9. Output pins 11 and 20 (UL1 and UL2) do not require use of the 2.5 V supply, CR1, R2, or C. R1 is the only load needed for these pins.

FIGURE 12. Logic load circuit.



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TABLE VIII. Input current, low level, test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Fig |
|---|---|---|---|---|---|---|---|---|---|---|
| H1 | 1D | 3,22 | 24 | 23 | Open | -0.40 | -0.03 | mA | 1-3 | 13 |
| H2 | 2D | 26,27 | 6 | 5 | | | | | | |
| H3 | 3D | 7,28 | 8 | 30 | | | | | | |
| H4 | 4D | 13,31 | 14 | 33 | | | | | | |
| H5 | 5D | 34,35 | 15 | 36 | | | | | | |
| H6 | 6D | 37,39 | 18 | 38 | | -0.40 | -0.03 | mA | 1-3 | 13 |
| H7 | 1S | Open | 23 | 3,22, 24 | | -0.80 | -0.06 | mA | 1-3 | 13 |
| H8 | 2S | | 5 | 6,26, 27 | | | | | | |
| H9 | 3S | | 30 | 7,8, 28 | | | | | | |
| H10 | 4S | | 33 | 13, 14,31 | | | | | | |
| H11 | 5S | | 36 | 15, 34,35 | | | | | | |
| H12 | 6S | Open | 38 | 18, 37,39 | | | | | | |
| H13 | 1C | 22 | 3 | 23,24 | | | | | | |
| H14 | 2C | 27 | 26 | 5,6 | | | | | | |
| H15 | 3C | 7 | 28 | 8,30 | | | | | | |
| H16 | 4C | 31 | 13 | 14,33 | | | | | | |
| H17 | 5C | 34 | 35 | 15,36 | Open | -0.80 | -0.06 | mA | 1-3 | 13 |



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TABLE VIII. Input current, low level, test - Continued.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Fig |
|---|---|---|---|---|---|---|---|---|---|---|
| H18 | 6C | 37 | 39 | 18,38 | Open | -0.80 | -0.06 | mA | 1-3 | 13 |
| H19 | 1R | 3,24 | 22 | 23 | | -1.20 | -0.06 | mA | 1-3 | 13 |
| H20 | 2R | 6,26 | 27 | 5 | | | | | | |
| H21 | 3R | 8,28 | 7 | 30 | | | | | | |
| H22 | 4R | 13,14 | 31 | 33 | | | | | | |
| H23 | 5R | 15,35 | 34 | 36 | | | | | | |
| H24 | 6R | 18,39 | 37 | 38 | Open | -1.20 | -0.06 | mA | 1-3 | 13 |

NOTES:

1. For test conditions H1-H24, set item P01-3 to 0.4 V ± 1 percent.

2. Do not isolate the ammeter M02 from earth ground.

3. With point B disconnected, item M02 should indicate zero current on the 200 uA range.



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TABLE IX Input current, high level, test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Fig |
|---|---|---|---|---|---|---|---|---|---|
| J1 | 1D | 3,23 | 24 | 22,5,6,26,27,30,8,28,7, 33,14,13,31,36,15,35,34, 38,18,39,37 | Open | 0 | 20 | uA | 13 |
| J2 | 2D | 5,26 | 6 | 23,24,3,22,27,30,8,28,7, 33,14,13,31,36,15,35,34, 38,18,39,37 | | | | | |
| J3 | 3D | 28,30 | 8 | 23,24,3,22,5,6,26,27,7, 33,14,13,31,36,15,35,34, 38,18,39,37 | | | | | |
| J4 | 4D | 13,33 | 14 | 23,24,3,22,27,30,8,28,5, 6,26,7,31,36,15,35,34, 38,18,39,37 | | | | | |
| J5 | 5D | 35,36 | 15 | 23,24,3,22,27,30,8,28,5, 6,26,7,33,14,13,31,34, 38,18,39,37 | | | | | |
| J6 | 6D | 38,39 | 18 | 23,24,3,22,27,30,8,28,5, 6,26,7,33,14,13,31,36, 15,35,34,37 | | 0 | 20 | uA | 13 |
| J7 | 1C | 23,24 | 3 | 22,5,6,26,27,30,8,28,7, 33,14,13,31,36,15,35,34, 38,18,39,37 | | 0 | 40 | uA | 13 |
| J8 | 2C | 5,6 | 26 | 23,24,3,22,27,30,8,28,7, 33,14,13,31,36,15,35,34, 38,18,39,37 | Open | 0 | 40 | uA | 13 |





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TABLE IX. Input current, high level, test - Continued.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Fig |
|---|---|---|---|---|---|---|---|---|---|
| J9 | 3C | 8,30 | 28 | 23,24,3,22,5,6,26,27,7, 33,14,13,31,36,15,35,34, 38,18,39,37 | Open | 0 | 40 | uA | 13 |
| J10 | 4C | 33,14 | 13 | 23,24,3,22,27,30,8,28,5, 6,26,7,31,36,15,35,34, 38,18,39,37 | | | | | |
| J11 | 5C | 36,15 | 35 | 23,24,3,22,27,30,8,28,5, 6,26,7,33,14,13,31,34, 38,18,39,37 | | | | | |
| J12 | 6C | 18,38 | 39 | 23,24,3,22,27,30,8,28,5, 6,26,7,33,14,13,31,36, 15,35,34,37 | Open | 0 | 40 | uA | 13 |
| J13 | 1S | 22,24 | 23 | 5,6,26,27,30,8,28,7, 33,14,13,31,36,15,35,34, 38,18,39,37 | 3 | 0 | 40 | uA | 13 |
| J14 | 2S | 6,27 | 5 | 23,24,3,22,30,8,28,7, 33,14,13,31,36,15,35,34, 38,18,39,37 | 26 | | | | |
| J15 | 3S | 7,8 | 30 | 23,24,3,22,5,6,26,27, 33,14,13,31,36,15,35,34, 38,18,39,37 | 28 | | | | |
| J16 | 4S | 14,31 | 33 | 23,24,3,22,27,30,8,28,5, 6,26,7,36,15,35,34,38, 18,39.37 | 13 | 0 | 40 | uA | 13 |





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TABLE IX. Input current, high level, test - Continued.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 + 5°C Initial limits Min | 25 + 5°C Initial limits Max | Units | Fig |
|---|---|---|---|---|---|---|---|---|---|
| J17 | 5S | 15,34 | 36 | 23,24,3,22,27,30,8,28,5, 6,26,7,33,14,13,31,38, 18,39,37 | 35 | 0 | 40 | uA | 13 |
| J18 | 6S | 18,37 | 38 | 23,24,3,22,27,30,8,28,5, 6,26,7,33,14,13,31,36, 15,35,34 | 39 | 0 | 40 | uA | 13 |
| J19 | 1R | 23 | 22 | 24,5,6,26,27,30,8,28,7, 33,14,13,31,36,15,35,34, 38,18,39,37 | 3 | 0 | 60 | uA | 13 |
| J20 | 2R | 5 | 27 | 23,24,3,22,6,30,8,28,7, 33,14,13,31,36,15,35,34, 38,18,39,37 | 26 | | | | |
| J21 | 3R | 30 | 7 | 23,24,3,22,5,6,26,27,8, 33,14,13,31,36,15,35,34, 38,18,39,37 | 28 | | | | |
| J22 | 4R | 33 | 31 | 23,24,3,22,27,30,8,28,5, 6,26,7,14,36,15,35,34, 38,18,39,37 | 13 | | | | |
| J23 | 5R | 36 | 34 | 23,24,3,22,27,30,8,28,5, 6,26,7,33,14,13,31,15, 38,18,39,37 | 35 | | | | |
| J24 | 6R | 38 | 37 | 23,24,3,22,27,30,8,28,5, 6,26,7,33,14,13,31,36, 15,35,34,18 | 39 | 0 | 60 | uA | 13 |





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NOTES:

1. For test conditions J1-J24, set item P01-3 to 2.7 V ± 1 percent.

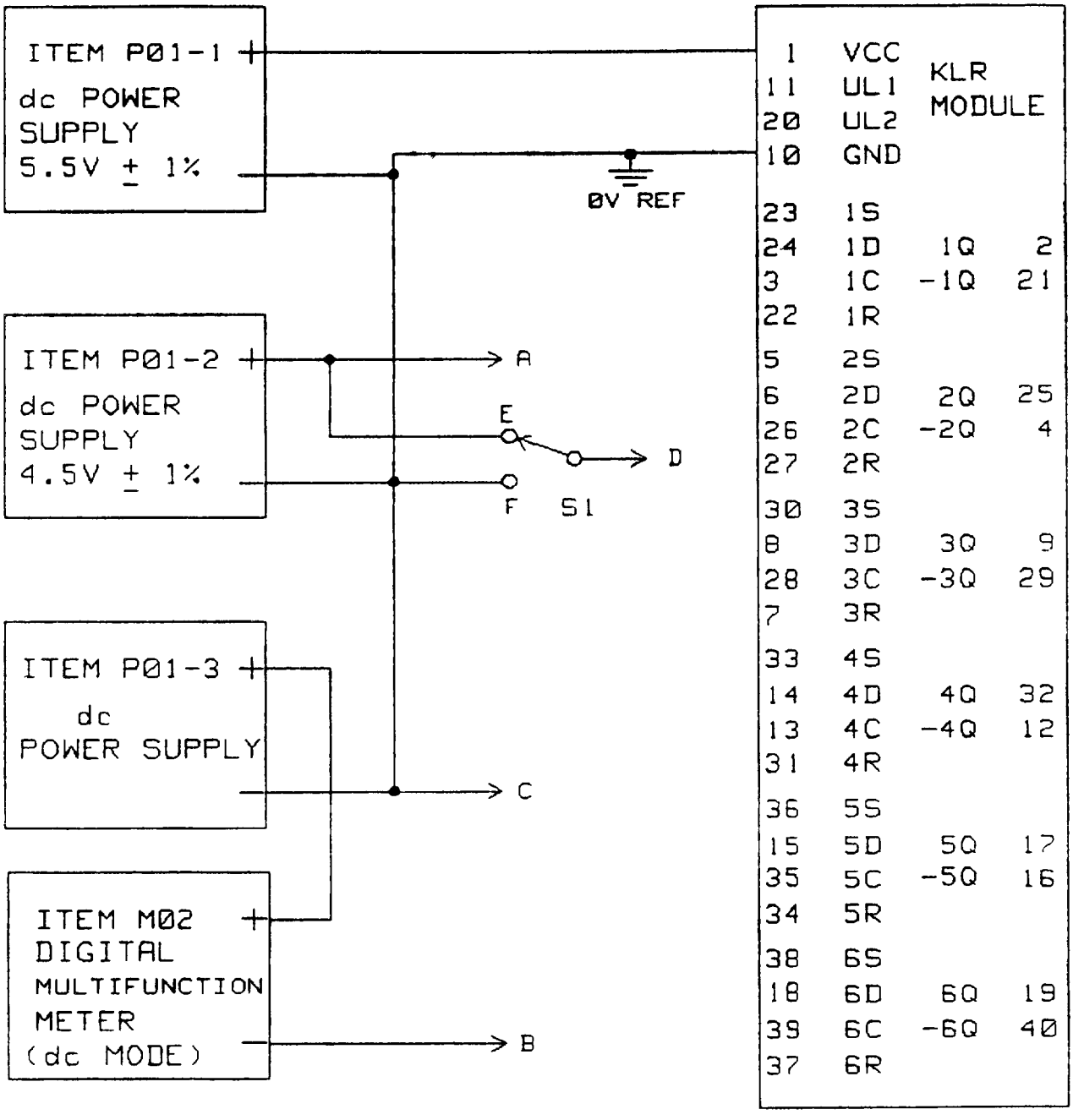
2. Do not isolate the ammeter M02 from earth ground.

3. For electrical requirements J13-J24, prior to making each measurement, set S1 in the E position, move S1 to the F position for a minimum of 100 nS, and return S1 to the E position, and make the required measurement.

4. With point B disconnected, item M02 should indicate zero current on the 200 uA range.



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NOTE:

1. S1 = DPST switch.

FIGURE 13. Input current, low and high level, test circuit for tables VIII and IX.



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TABLE X. Power supply current test.

| Test cond | Parameter | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Fig |
|---|---|---|---|---|---|---|---|---|
| K1 | Icc (low level) | C | D | 1 | 36 | mA | 1 | 14 |
| K2 | Icc (high level) | D | C | 1 | 36 | mA | 1 | 14 |

NOTE:

1. Do not isolate the chassis of ammeter X01 from earth ground.



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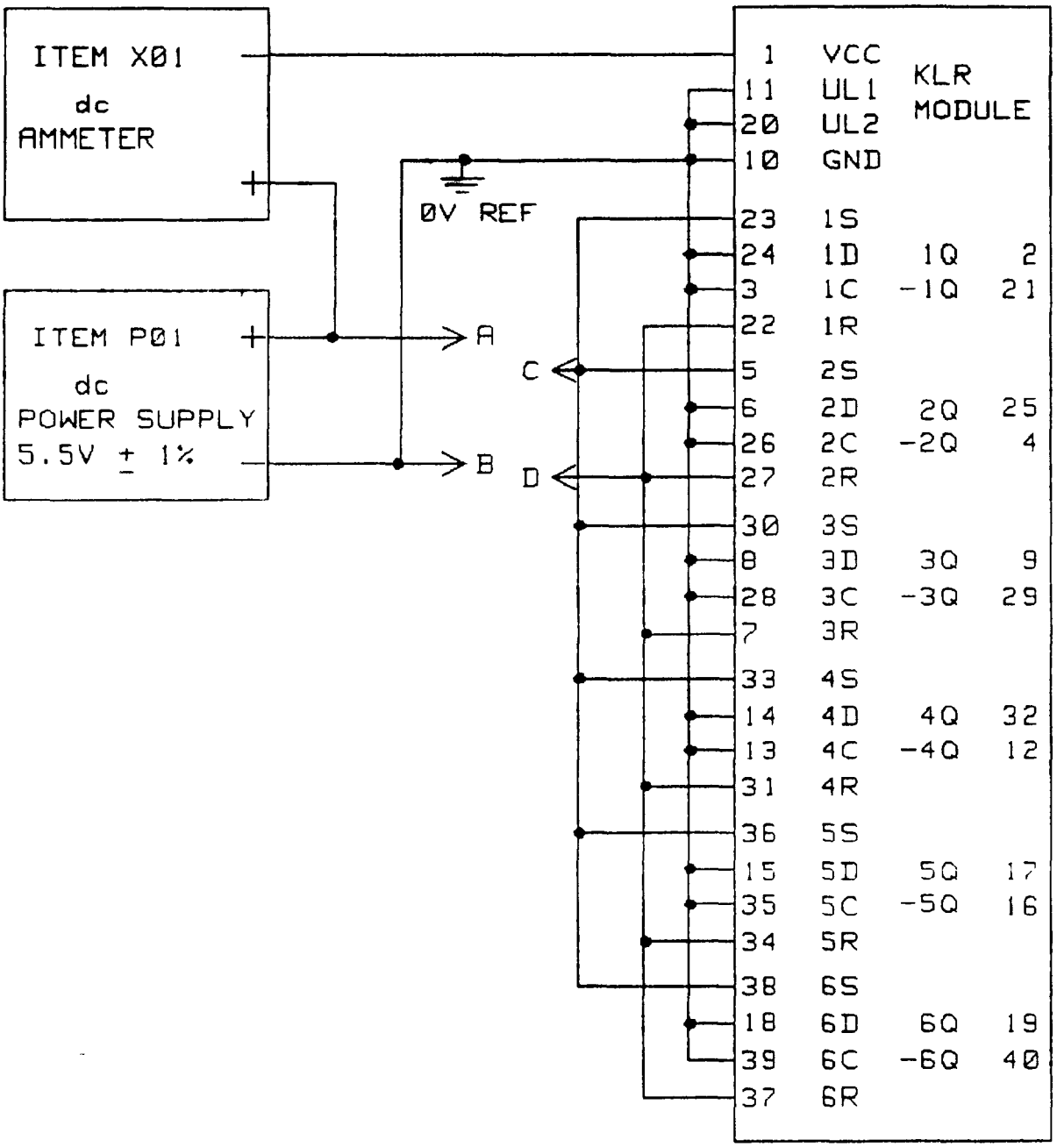



FIGURE 14. Power supply current test circuit for table X.



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TABLE XI. Propagation delay time (tPHL) test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Fig |
|---|---|---|---|---|---|---|---|---|---|
| L1 | 1Q | 3 | 24 | 2 | 5 | 57 | nS | 1,2 | 15,16 |
| L2 | -1Q | 3 | 24 | 21 | | | | | |
| L3 | 2Q | 26 | 6 | 25 | | | | | |
| L4 | -2Q | 26 | 6 | 4 | | | | | |
| L5 | 3Q | 28 | 8 | 9 | | | | | |
| L6 | -3Q | 28 | 8 | 29 | | | | | |
| L7 | 4Q | 13 | 14 | 32 | | | | | |
| L8 | -4Q | 13 | 14 | 12 | | | | | |
| L9 | 5Q | 35 | 15 | 17 | | | | | |
| L10 | -5Q | 35 | 15 | 16 | | | | | |
| L11 | 6Q | 39 | 18 | 19 | | | | | |
| L12 | -6Q | 39 | 18 | 40 | 5 | 57 | nS | 1,2 | 15,16 |

NOTES:

1. Load outputs as shown on figure 12.

2. Before performing this test, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.5 V for the high level.



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TABLE XII. Propagation delay time (tPLH) test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Fig |
|---|---|---|---|---|---|---|---|---|---|
| M1 | 1Q | 3 | 24 | 2 | 5 | 46 | nS | 1,2 | 15,16 |
| M2 | -1Q | 3 | 24 | 21 | | | | | |
| M3 | 2Q | 26 | 6 | 25 | | | | | |
| M4 | -2Q | 26 | 6 | 4 | | | | | |
| M5 | 3Q | 28 | 8 | 9 | | | | | |
| M6 | -3Q | 28 | 8 | 29 | | | | | |
| M7 | 4Q | 13 | 14 | 32 | | | | | |
| M8 | -4Q | 13 | 14 | 12 | | | | | |
| M9 | 5Q | 35 | 15 | 17 | | | | | |
| M10 | -5Q | 35 | 15 | 16 | | | | | |
| M11 | 6Q | 39 | 18 | 19 | | | | | |
| M12 | -6Q | 39 | 18 | 40 | 5 | 46 | nS | 1,2 | 15,16 |

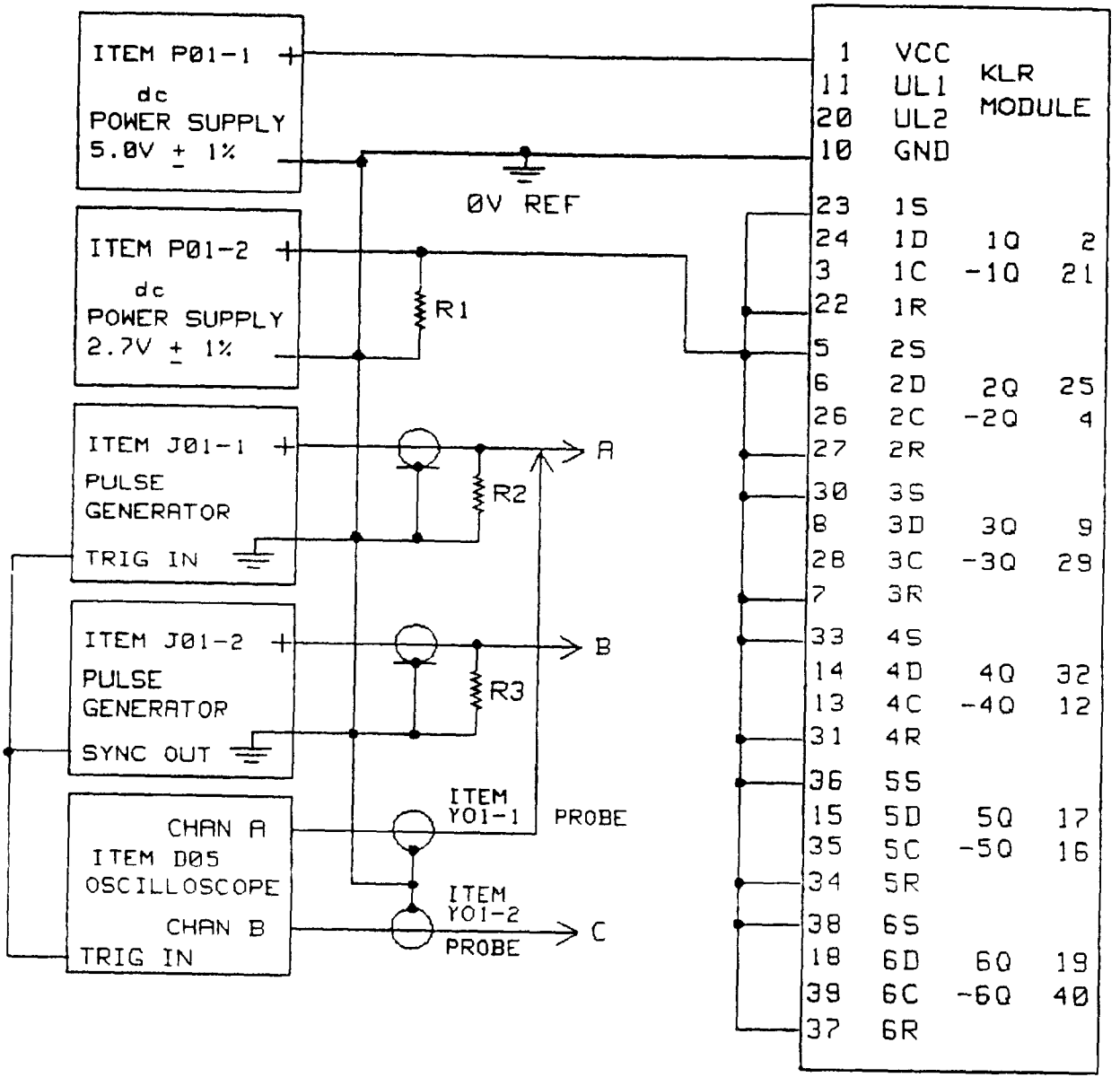
NOTES:

1. Load outputs as shown on figure 12.

2. Before performing this test, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.5 V for the high level.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
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NOTE:

1. R1 = R2 = R3 = 51 Ohms ± 5 percent, 1 W.

FIGURE 15. Propagation delay time test circuit for tables XI and XII.



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Check the source to verify that this is the current version before use.

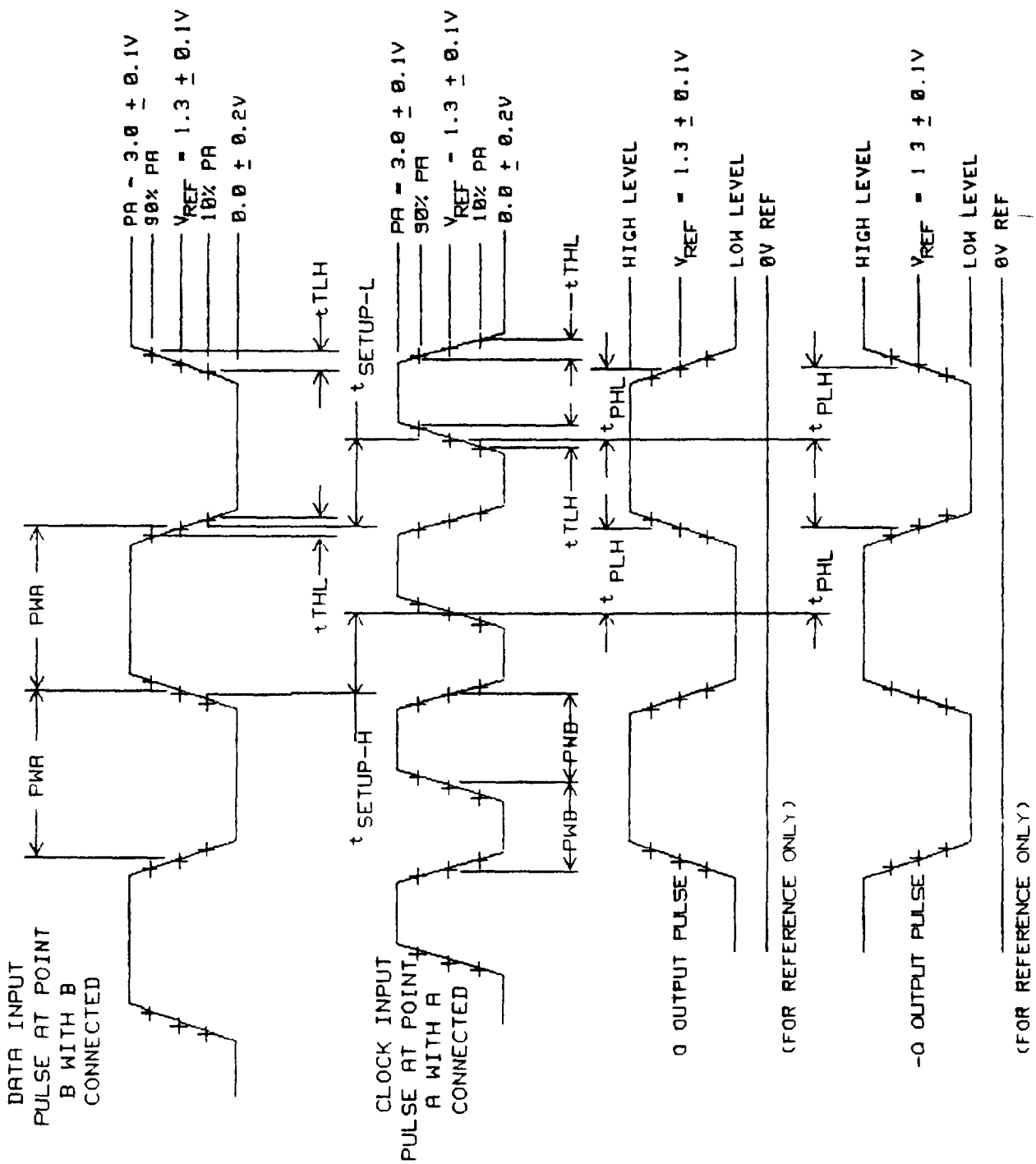



FIGURE 16. Propagation delay time test waveforms for tables XI and XII.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
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NOTE:

1. Input pulse parameters:

   PWA = 500 ± 50 nS.
   PWB = 250 ± 25 nS.
   tTLH = tTHL = 10 ± 2 nS.
   tSETUP-H = tSETUP-L = 125 ± 12 nS.



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TABLE XIII. Propagation delay time (tPHL) test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Figs |
|---|---|---|---|---|---|---|---|---|---|---|
| N1 | 1Q | 23 | 22,C | B | 2 | 5 | 57 | nS | 1,2 | 17,18 |
| N2 | -1Q | 23,C | 22 | A | 21 | | | | | |
| N3 | 2Q | 5 | 27,C | B | 25 | | | | | |
| N4 | -2Q | 5,C | 27 | A | 4 | | | | | |
| N5 | 3Q | 30 | 7,C | B | 9 | | | | | |
| N6 | -3Q | 30,C | 7 | A | 29 | | | | | |
| N7 | 4Q | 33 | 31,C | B | 32 | | | | | |
| N8 | -4Q | 33,C | 31 | A | 12 | | | | | |
| N9 | 5Q | 36 | 34,C | B | 17 | | | | | |
| N10 | -5Q | 36,C | 34 | A | 16 | | | | | |
| N11 | 6Q | 38 | 37,C | B | 19 | | | | | |
| N12 | -6Q | 38,C | 37 | A | 40 | 5 | 57 | nS | 1,2 | 17,18 |

NOTES:

1. Load outputs as shown on figure 12.

2. Before performing this test, the output levels shall be verified to be ≤ 0.4 V for the low level and ≥ 2.5 V for the high level.



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TABLE XIV. Propagation delay time (tPLH) test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Figs |
|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 1Q | 23,C | 22 | A | 2 | 5 | 46 | nS | 1,2 | 17,18 |
| P2 | -1Q | 23 | 22,C | B | 21 | | | | | |
| P3 | 2Q | 5,C | 27 | A | 25 | | | | | |
| P4 | -2Q | 5 | 27,C | B | 4 | | | | | |
| P5 | 3Q | 30,C | 7 | A | 9 | | | | | |
| P6 | -3Q | 30 | 7,C | B | 29 | | | | | |
| P7 | 4Q | 33,C | 31 | A | 32 | | | | | |
| P8 | -4Q | 33 | 31,C | B | 12 | | | | | |
| P9 | 5Q | 36,C | 34 | A | 17 | | | | | |
| P10 | -5Q | 36 | 34,C | B | 16 | | | | | |
| P11 | 6Q | 38,C | 37 | A | 19 | | | | | |
| P12 | -6Q | 38 | 37,C | B | 40 | 5 | 46 | nS | 1,2 | 17,18 |

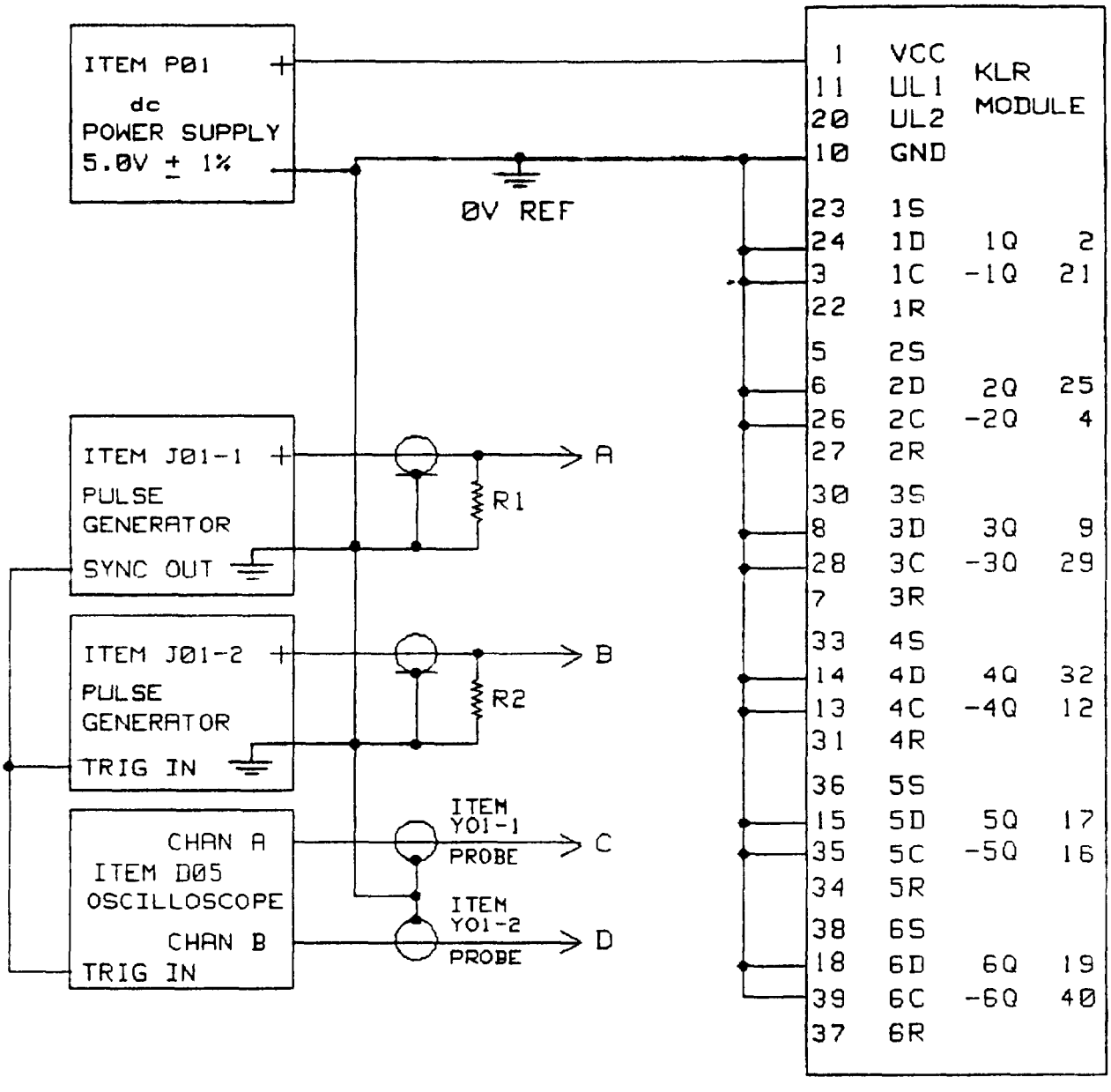
NOTES:

1. Load outputs as shown on figure 12.

2. Before performing this test, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.5 V for the high level.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.




NOTE:

1. R1 = R2 = 51 Ohms ± 5 percent, 1 W.

FIGURE 17. Propagation delay time test circuit for tables XIII and XIV.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

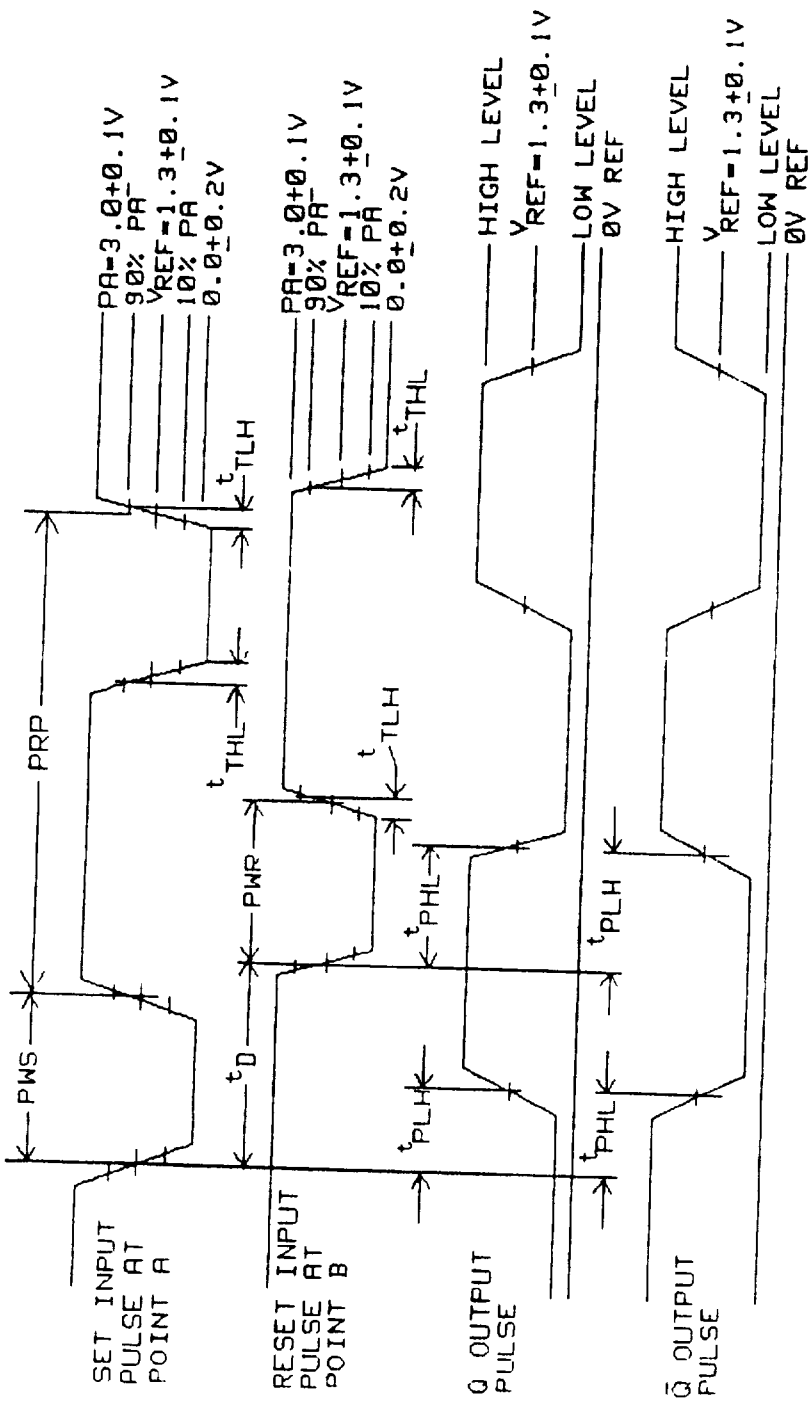



FIGURE 18. Propagation delay time (tPLH and tPHL) test circuit waveforms for tables XIII and XIV.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES:

1. Adjust item J01-1 and J01-2 to provide input pulse parameters as follows:

PWS = PWR = 35 ± 2 nS.
PRP = 200 ± 10 nS.
tD = 100 ± 5 nS.
tTLH = tTHL = 10 ± 2 nS.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
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TABLE XV. Setup time test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Figs |
|---|---|---|---|---|---|---|---|---|---|
| Q1 | 1Q | 3 | 24 | 2 | 900 | 1100 | nS | 1,2 | 19,20 |
| Q2 | 2Q | 26 | 6 | 25 | | | | | |
| Q3 | 3Q | 28 | 8 | 9 | | | | | |
| Q4 | 4Q | 13 | 14 | 32 | | | | | |
| Q5 | 5Q | 35 | 15 | 17 | | | | | |
| Q6 | 6Q | 39 | 18 | 19 | | | | | |
| Q7 | -1Q | 3 | 24 | 21 | | | | | |
| Q8 | -2Q | 26 | 6 | 4 | | | | | |
| Q9 | -3Q | 28 | 8 | 29 | | | | | |
| Q10 | -4Q | 13 | 14 | 12 | | | | | |
| Q11 | -5Q | 35 | 15 | 16 | | | | | |
| Q12 | -6Q | 39 | 18 | 40 | 900 | 1100 | nS | 1,2 | 19,20 |

NOTES:

1. Load outputs as shown on figure 12.

2. Before performing this test, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.5 V for the high level.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
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TABLE XVI. Hold time test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Figs |
|---|---|---|---|---|---|---|---|---|---|
| R1 | 1Q | 3 | 24 | 2 | 900 | 1100 | nS | 1,2 | 19,21 |
| R2 | 2Q | 26 | 6 | 25 | | | | | |
| R3 | 3Q | 28 | 8 | 9 | | | | | |
| R4 | 4Q | 13 | 14 | 32 | | | | | |
| R5 | 5Q | 35 | 15 | 17 | | | | | |
| R6 | 6Q | 39 | 18 | 19 | | | | | |
| R7 | -1Q | 3 | 24 | 21 | | | | | |
| R8 | -2Q | 26 | 6 | 4 | | | | | |
| R9 | -3Q | 28 | 8 | 29 | | | | | |
| R10 | -4Q | 13 | 14 | 12 | | | | | |
| R11 | -5Q | 35 | 15 | 16 | | | | | |
| R12 | -6Q | 39 | 18 | 40 | 900 | 1100 | nS | 1,2 | 19,21 |

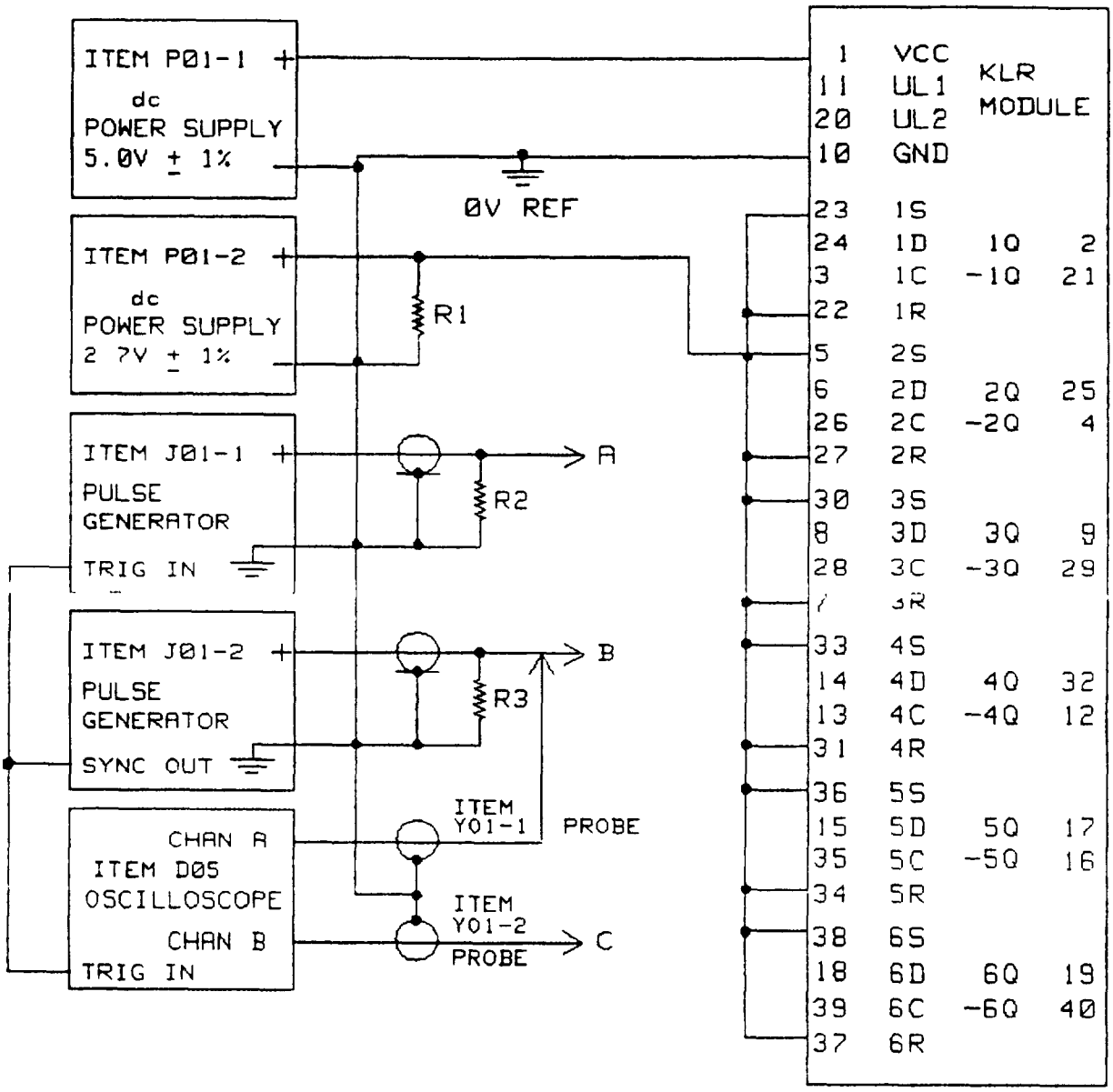
NOTES:

1. Load outputs as shown on figure 12.

2. Before performing this test, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.5 V for the high level.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.




NOTE:

1. R1 = R2 = R3 = 51 Ohms ± 5 percent, 1 W.

FIGURE 19. Setup time and hold time test circuit for tables XV and XVI.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

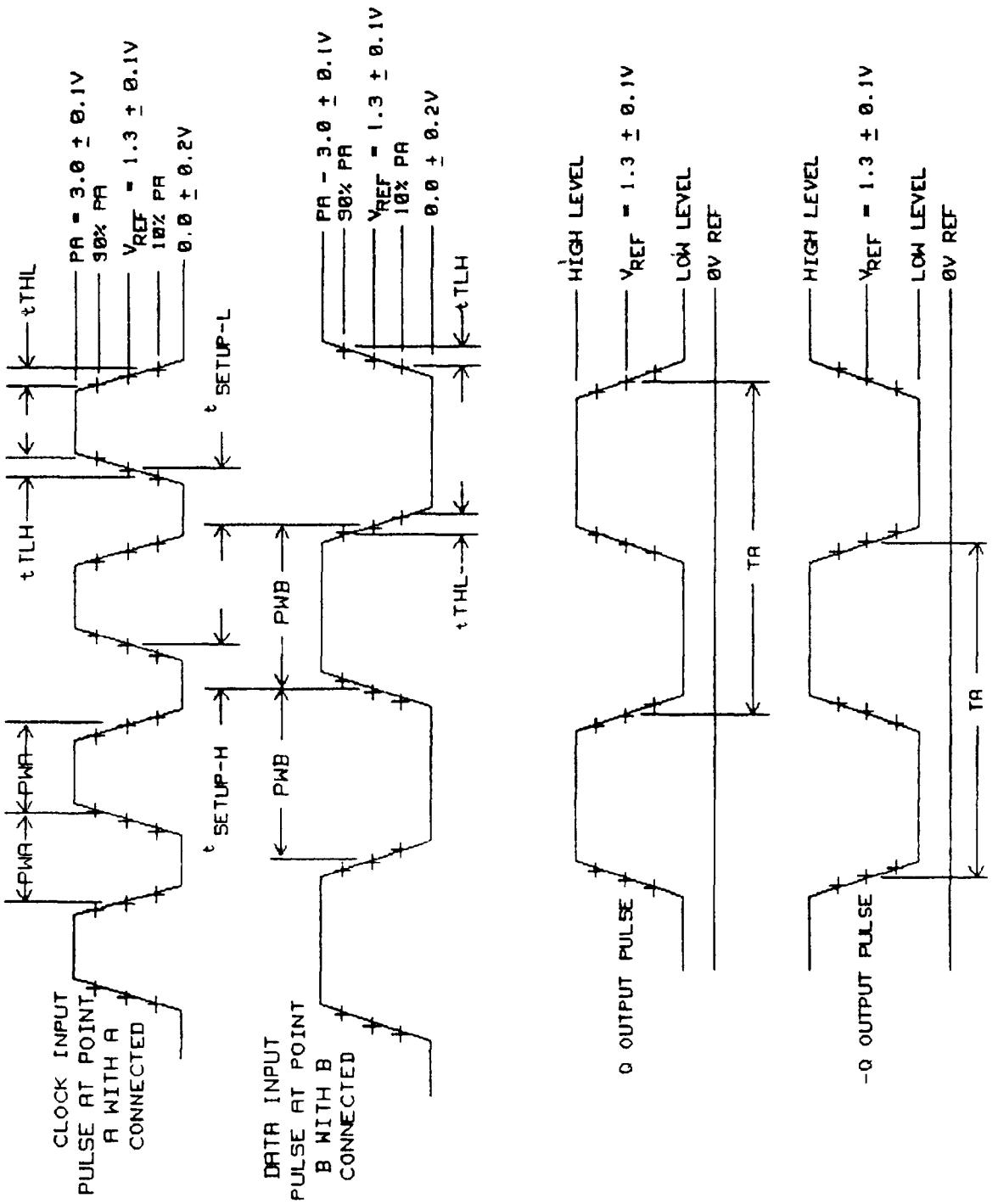



FIGURE 20. Setup time test waveforms for table XV.



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NOTES:

1. Input pulse parameters:

   PWA = 250 ± 25 nS.
   PWB = 500 ± 50 nS.
   tTLH = tTHL = 10 ± 2 nS.
   tSETUP-H = tSETUP-L = 20 +2, -0 nS.



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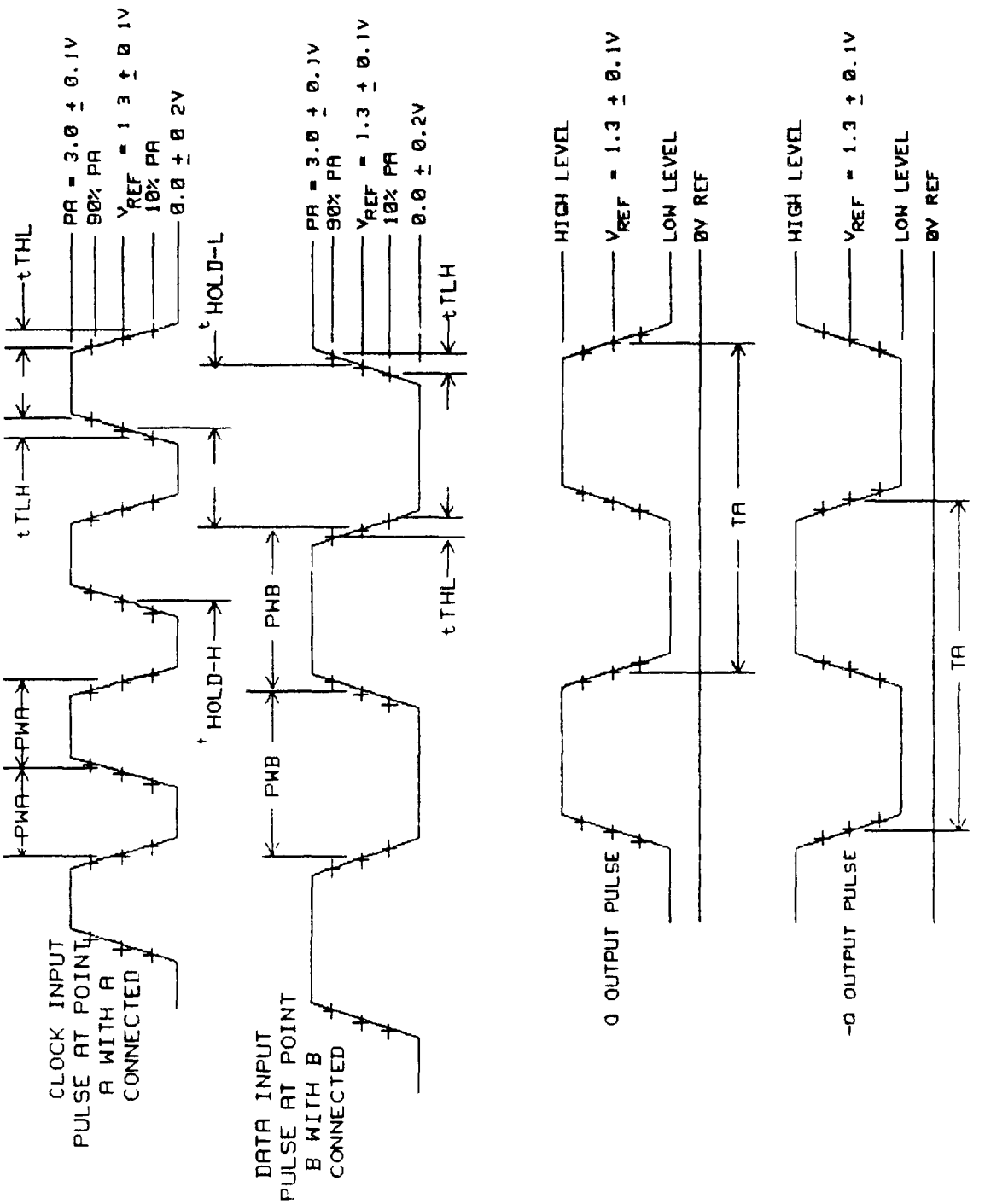



FIGURE 21. Hold time test waveforms for table XVI.



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NOTES:

1. Input pulse parameters:

   PWA = 250 ± 25 nS.
   PWB = 500 ± 50 nS.
   tTLH = tTHL = 10 ± 2 nS.
   tHOLD-H = tHOLD-L = 5 +2, -0 nS.



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TABLE XVII. Maximum clock frequency/minimum clock pulse width test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Figs |
|---|---|---|---|---|---|---|---|---|---|
| S1 | 1Q | 3 | 24 | 2 | 92 | 108 | nS | 1,2 | 22,23 |
| S2 | 2Q | 26 | 6 | 25 | | | | | |
| S3 | 3Q | 28 | 8 | 9 | | | | | |
| S4 | 4Q | 13 | 14 | 32 | | | | | |
| S5 | 5Q | 35 | 15 | 17 | | | | | |
| S6 | 6Q | 39 | 18 | 19 | | | | | |
| S7 | -1Q | 3 | 24 | 21 | | | | | |
| S8 | -2Q | 26 | 6 | 4 | | | | | |
| S9 | -3Q | 28 | 8 | 29 | | | | | |
| S10 | -4Q | 13 | 14 | 12 | | | | | |
| S11 | -5Q | 35 | 15 | 16 | | | | | |
| S12 | -6Q | 39 | 18 | 40 | 92 | 108 | nS | 1,2 | 22,23 |

NOTES:

1. Load outputs as shown on figure 12.

2. Before performing this test, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.5 V for the high level.



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TABLE XVIII. Threshold clock test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Notes | Figs |
|---|---|---|---|---|---|---|---|---|---|
| T1 | 1Q | 3 | 24 | 2 | 900 | 1100 | nS | 1,2 | 22,23 |
| T2 | 2Q | 26 | 6 | 25 | | | | | |
| T3 | 3Q | 28 | 8 | 9 | | | | | |
| T4 | 4Q | 13 | 14 | 32 | | | | | |
| T5 | 5Q | 35 | 15 | 17 | | | | | |
| T6 | 6Q | 39 | 18 | 19 | | | | | |
| T7 | -1Q | 3 | 24 | 21 | | | | | |
| T8 | -2Q | 26 | 6 | 4 | | | | | |
| T9 | -3Q | 28 | 8 | 29 | | | | | |
| T10 | -4Q | 13 | 14 | 12 | | | | | |
| T11 | -5Q | 35 | 15 | 16 | | | | | |
| T12 | -6Q | 39 | 18 | 40 | 900 | 1100 | nS | 1,2 | 22,23 |

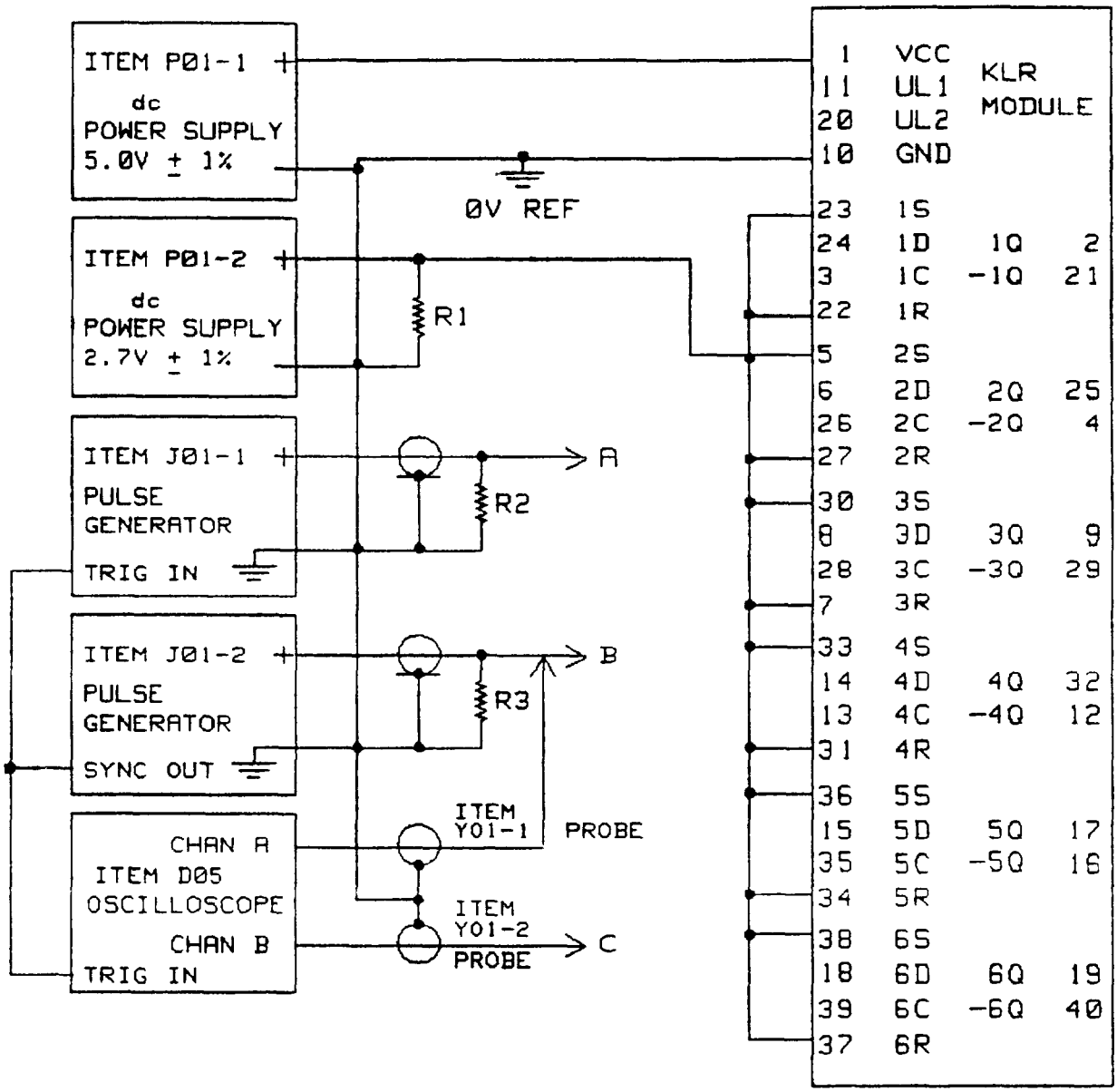
NOTES:

1. Load outputs as shown on figure 12.

2. Before performing this test, the output levels shall be verified to be $\leq$ 0.4 V for the low level and $\geq$ 2.5 V for the high level.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
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NOTES:

1. R1 = R2 = R3 = 51 Ohms ± 5 percent, 1 W.

FIGURE 22. Maximum clock frequency/minimum clock pulse width and threshold clock test for tables XVII and XVIII.



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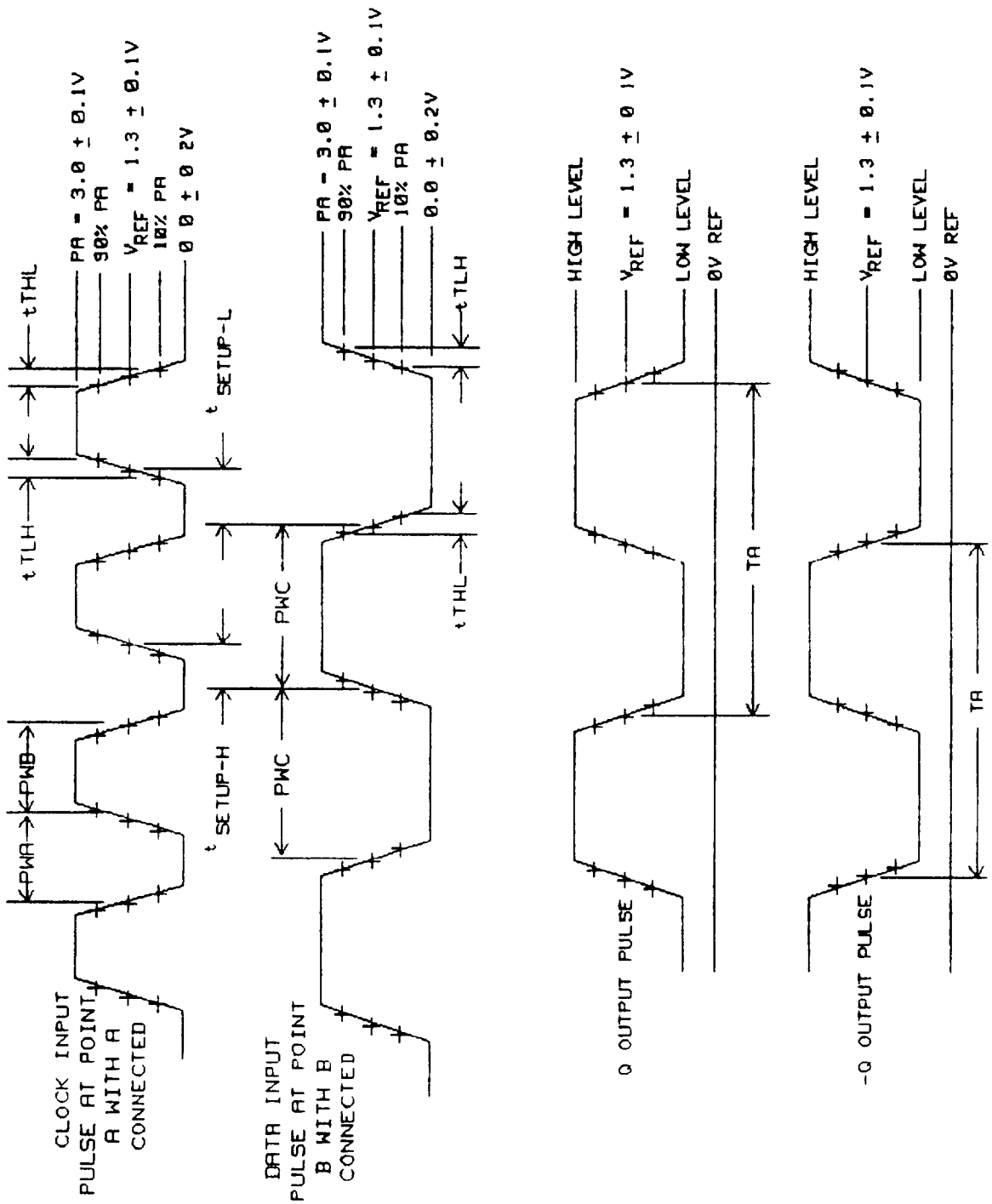



FIGURE 23. Maximum clock frequency/minimum clock pulse width test circuit waveforms for table XVII.



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NOTES:

1. Input pulse parameters:

PWA = 20 ± 2 nS.
PWB = 30 ± 4 nS.
PWC = 50 ± 4 nS.
tTLH = tTHL = 10 ± 2 nS.
tSETUP-H = tSETUP-L = 30 ± 1.5 nS.



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Check the source to verify that this is the current version before use.

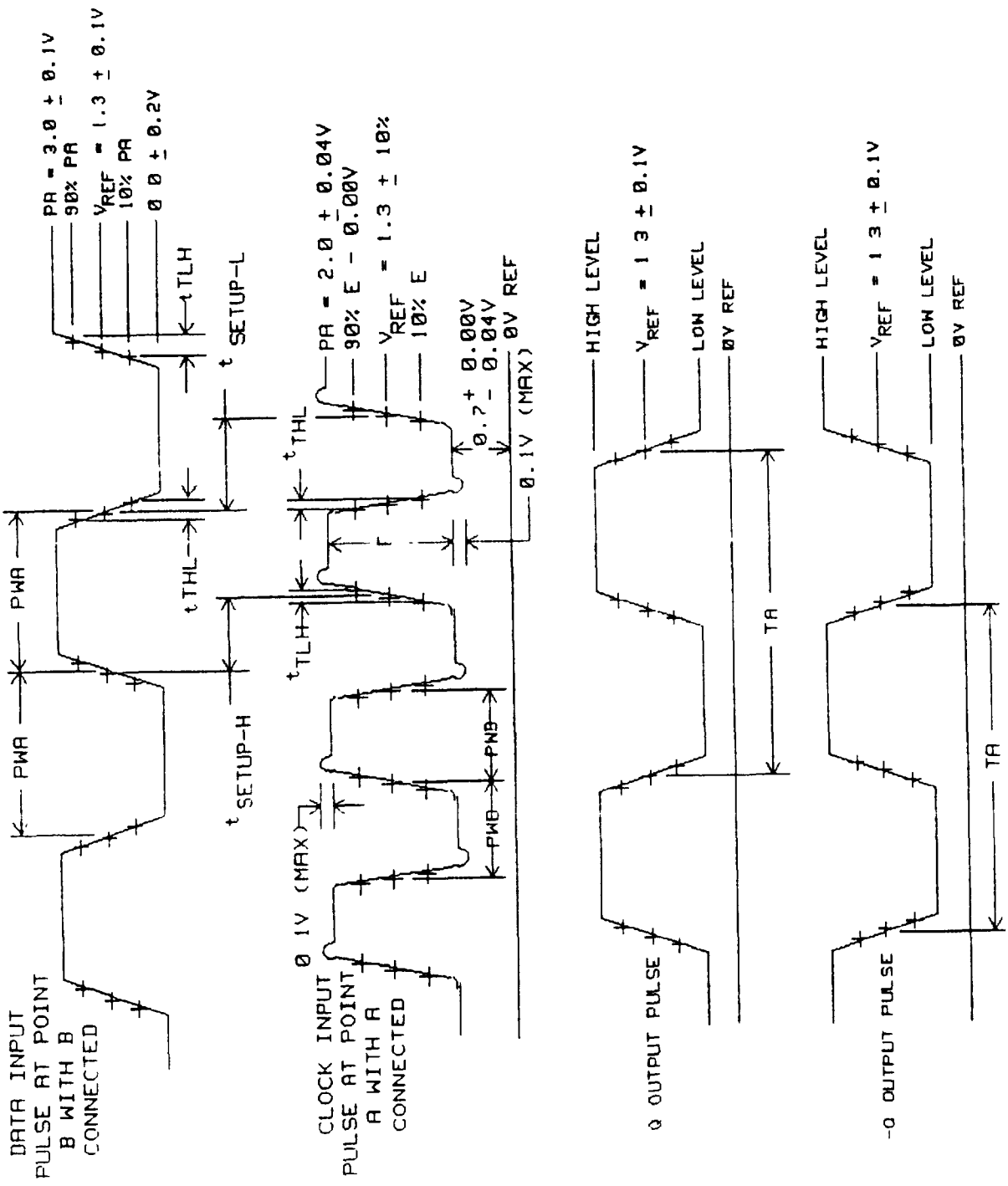



FIGURE 24. Threshold clock test circuit waveforms for table XVIII.



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NOTES:

1. Input pulse parameters:

   PWA = 500 ± 50 nS.
   PWB = 250 ± 25 nS.
   tTLH = tTHL = 25 ± 2.5 nS.
   tSETUP-H = tSETUP-L = 125 ± 12.5 nS.



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TABLE XIX. Output crosstalk, low and high level, test.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Figs |
|---|---|---|---|---|---|---|---|---|
| U1 | 1Q Low level | 5,7,23,27,30,31, 33,34,36,37,38 | 22 | 2 | 0 | 400 | mV | 25,26 |
| U2 | -1Q High level | 5,7,23,27,30,31, 33,34,36,37,38 | 22 | 21 | | | | |
| U3 | -1Q Low level | 5,7,22,27,30,31, 33,34,36,37,38 | 23 | 21 | | | | |
| U4 | 1Q High level | 5,7,22,27,30,31, 33,34,36,37,38 | 23 | 2 | | | | |
| U5 | 2Q Low level | 5,7,22,23,30,31, 33,34,36,37,38 | 27 | 25 | | | | |
| U6 | -2Q High level | 5,7,22,23,30,31, 33,34,36,37,38 | 27 | 4 | | | | |
| U7 | -2Q Low level | 7,22,23,27,30,31, 33,34,36,37,38 | 5 | 4 | | | | |
| U8 | 2Q High level | 7,22,23,27,30,31, 33,34,36,37,38 | 5 | 25 | | | | |
| U9 | 3Q Low level | 5,22,23,27,30,31, 33,34,36,37,38 | 7 | 9 | 0 | 400 | mV | 25,26 |





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TABLE XIX. Output crosstalk, low level and high level, test - Continued.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Figs |
|---|---|---|---|---|---|---|---|---|
| U10 | -3Ω High level | 5,22,23,27,30,31, 33,34,36,37,38 | 7 | 29 | 0 | 400 | mV | 25,26 |
| U11 | -3Ω Low level | 5,7,22,23,27,31, 33,34,36,37,38 | 30 | 29 | | | | |
| U12 | 3Ω High level | 5,7,22,23,27,31, 33,34,36,37,38 | 30 | 9 | | | | |
| U13 | 4Ω Low level | 5,7,22,23,27,30, 33,34,36,37,38 | 31 | 33 | | | | |
| U14 | -4Ω High level | 5,7,22,23,27,30, 33,34,36,37,38 | 31 | 13 | | | | |
| U15 | -4Ω Low level | 5,7,22,23,27,30, 31,34,36,37,38 | 33 | 13 | | | | |
| U16 | 4Ω High level | 5,7,22,23,27,30, 31,34,36,37,38 | 33 | 33 | | | | |
| U17 | 5Ω Low level | 5,7,22,23,27,30, 31,33,36,37,38 | 34 | 17 | | | | |
| U18 | -5Ω High level | 5,7,22,23,27,30, 31,33,36,37,38 | 34 | 16 | 0 | 400 | mV | 25,26 |



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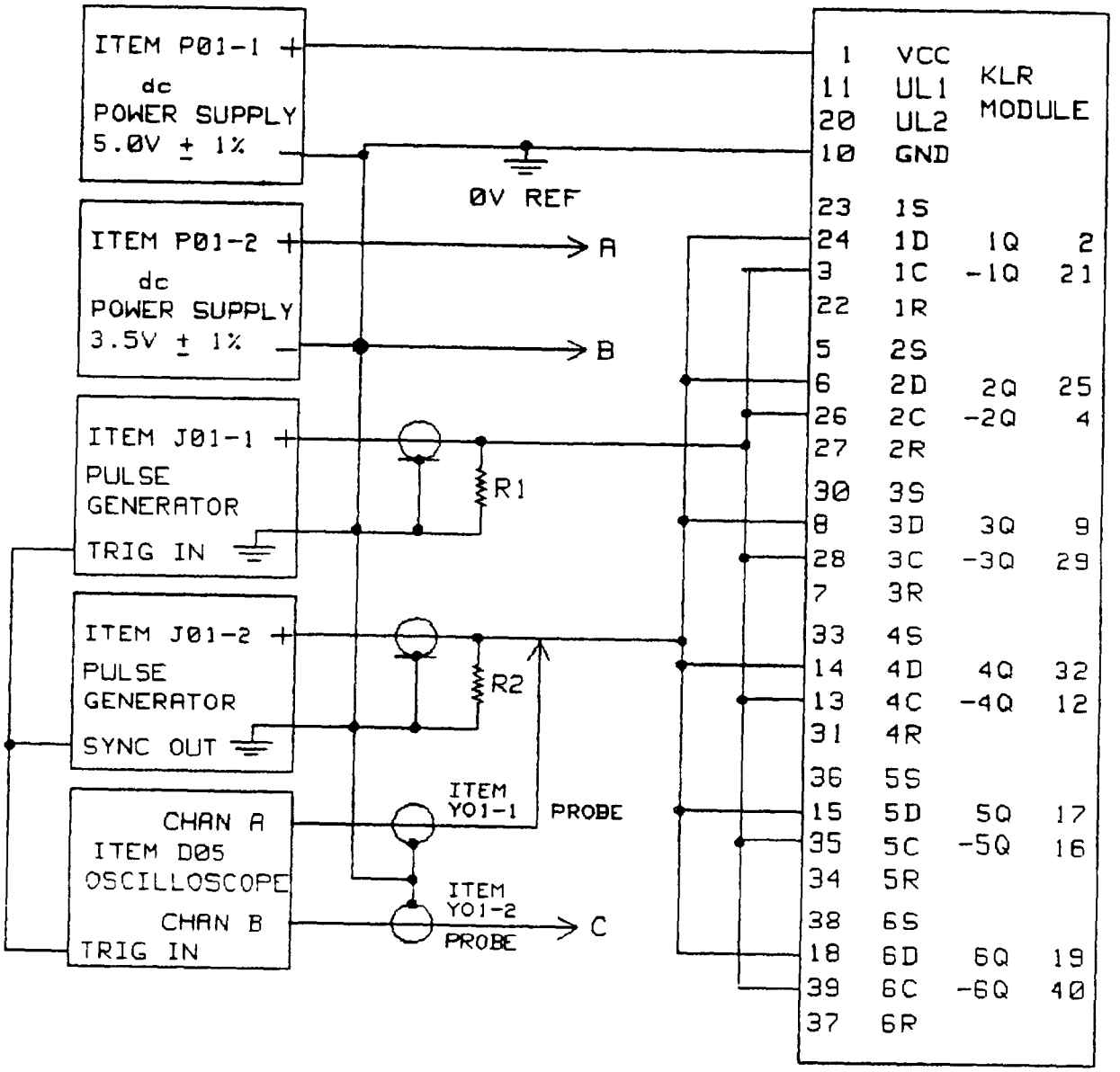
TABLE XIX Output crosstalk, low level and high level, test - Continued.

| Test cond | Signal name | Connect A to pin | Connect B to pin | Connect C to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Figs |
|---|---|---|---|---|---|---|---|---|
| U19 | -5Q Low level | 5,7,22,23,27,30, 31,33,34,37,38 | 36 | 16 | 0 | 400 | mV | 25,26 |
| U20 | 5Q High level | 5,7,22,23,27,30, 31,33,34,37,38 | 36 | 17 | | | | |
| U21 | 6Q Low level | 5,7,22,23,27,30, 31,33,34,36,38 | 37 | 19 | | | | |
| U22 | -6Q High level | 5,7,22,23,27,30, 31,33,34,36,38 | 37 | 40 | | | | |
| U23 | -6Q Low level | 5,7,22,23,27,30, 31,33,34,36,37 | 38 | 40 | | | | |
| U24 | 6Q High level | 5,7,22,23,27,30, 31,33,34,36,37 | 38 | 19 | 0 | 400 | mV | 25,26 |





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NOTES:

1. R1 = R2 = 51 Ohms ± 5 percent, 1 W.
2. Load outputs as shown on figure 12.

FIGURE 25. Output crosstalk, low and high level, test circuit for table XIX.



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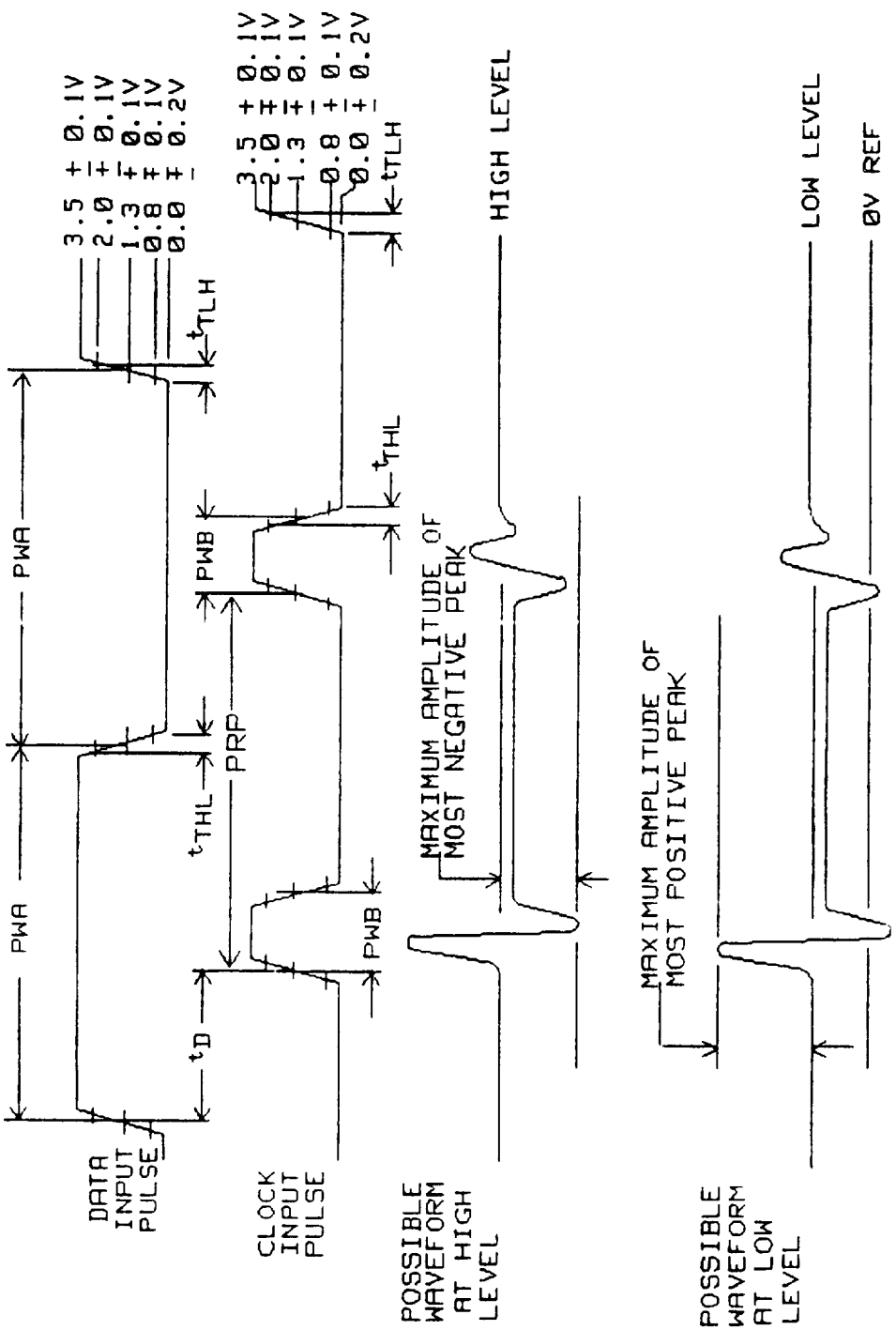



FIGURE 26. Output crosstalk, low and high level, test circuit waveforms for table XIX.



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NOTE:

1. Input pulse parameters:

   PWA = 500 ± 50 nS.
   PWB = 100 ± 10 nS.
   PRP (Clock) = 500 ± 50 nS.
   tTLH = tTHL = 3.5 ± 1.5 nS (verified at each input).
   tD = 200 ± 20 nS.



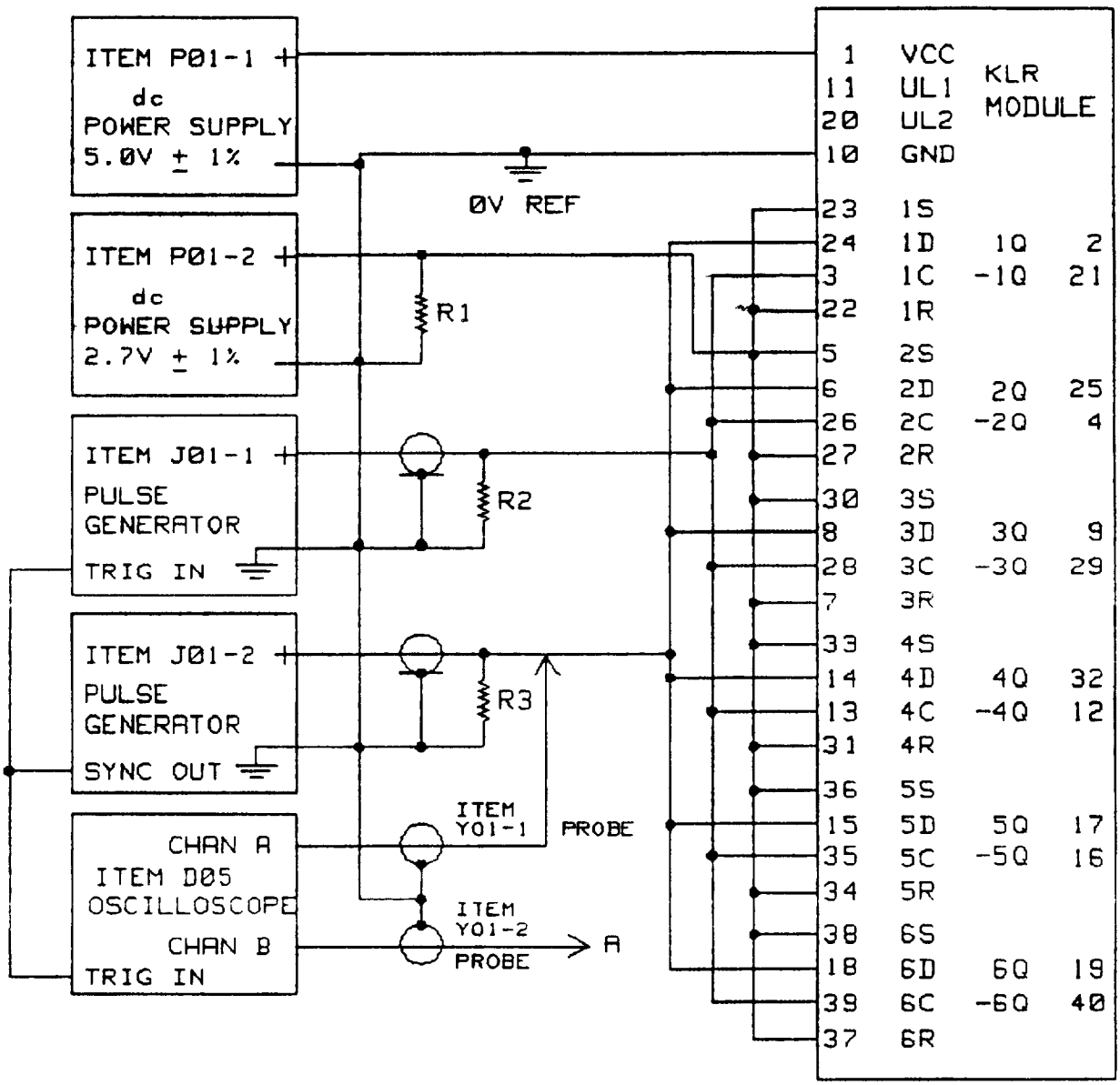
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
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TABLE XX. Power supply transient amplitude test.

| Test cond | Parameter | 25 ± 5°C Initial limits | | Units | Figs |
|---|---|---|---|---|---|
| | | Min | Max | | |
| V | PSTA | 0 | 400 | mV | 27,28 |



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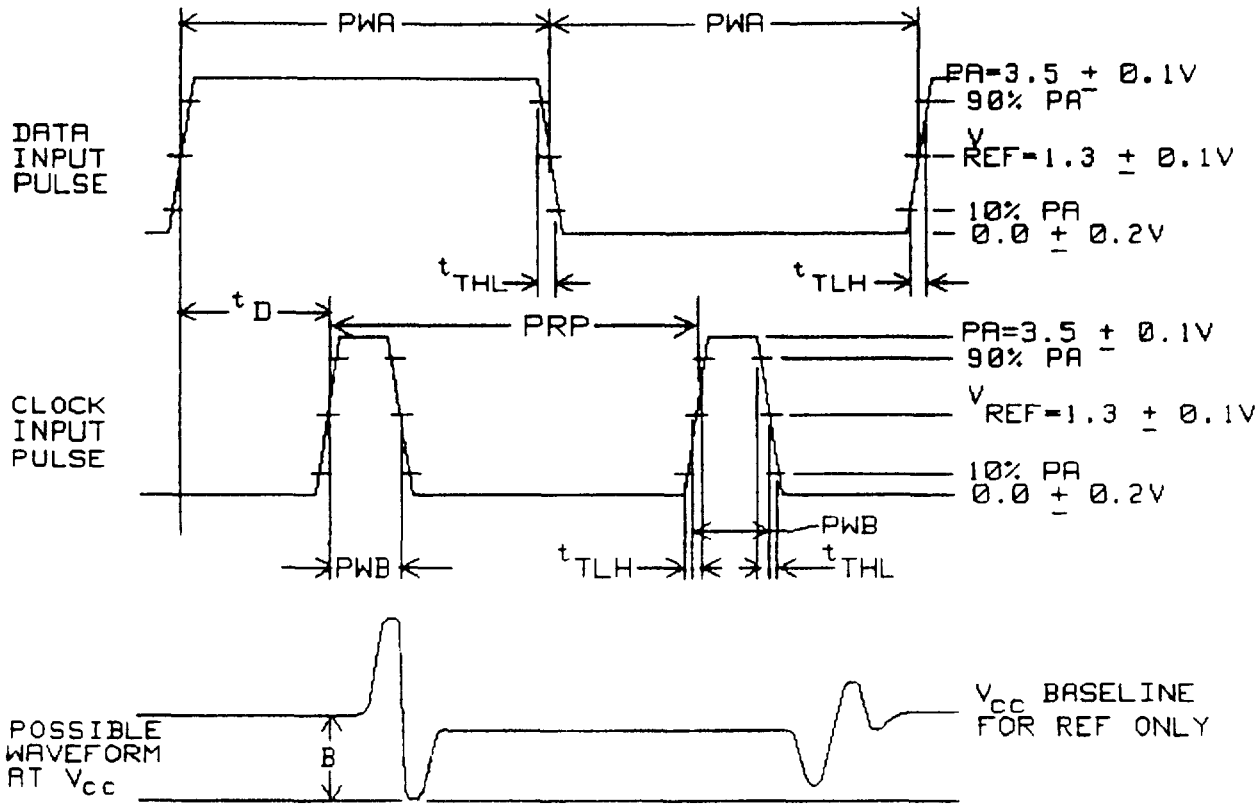

NOTE:

1. R1 = R2 = R3 = 51 Ohms ± 5 percent, 1 W.

FIGURE 27. Power supply transient amplitude test circuit for table XX.



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NOTES:

1. Input pulse parameters:

   PWA = 500 ± 50 nS.
   PWB = 100 ± 10 nS.
   PRP (Clock) = 500 ± 50 nS.
   tTLH = tTHL = 10 ± 1 nS.
   tD = 200 ± 20 nS.

2. Connect point A as close as possible to pin 1 on the module. Observe power supply transient on item D05; typical waveforms are shown above.

3. B = measurement from the baseline to the most negative excursion of the waveform.

FIGURE 28. Power supply transient amplitude test circuit waveforms for table XX.



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TABLE XXI. Power supply and input capacitance test.

| Test cond | Signal name (parameter) | Connect A to pin | Connect B to pin | 25 ± 5°C Initial limits Min | 25 ± 5°C Initial limits Max | Units | Fig |
|---|---|---|---|---|---|---|---|
| W | PSFC | Open | 1 | 3.6 | 6.0 | uF | 29 |
| X1 | 1C (Cin) | 3 | Open | 2.0 | 18.0 | pF | 29 |
| X2 | 1S (Cin) | 23 | | | | | |
| X3 | 1R (Cin) | 22 | | | | | |
| X4 | 1D (Cin) | 24 | | | | | |
| X5 | 2S (Cin) | 5 | | | | | |
| X6 | 2D (Cin) | 6 | | | | | |
| X7 | 2R (Cin) | 27 | | | | | |
| X8 | 2C (Cin) | 26 | | | | | |
| X9 | 3D (Cin) | 8 | | | | | |
| X10 | 3C (Cin) | 28 | | | | | |
| X11 | 3S (Cin) | 30 | | | | | |
| X12 | 3R (Cin) | 7 | | | | | |
| X13 | 4C (Cin) | 13 | | | | | |
| X14 | 4D (Cin) | 14 | | | | | |
| X15 | 4S (Cin) | 33 | | | | | |
| X16 | 4R (Cin) | 31 | | | | | |
| X17 | 5D (Cin) | 15 | | | | | |
| X18 | 5C (Cin) | 35 | | | | | |
| X19 | 5S (Cin) | 36 | Open | 2.0 | 18.0 | pF | 29 |



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TABLE XXI. Power supply and input capacitance test - Continued.

| Test cond | Signal name (parameter) | Connect | | 25 ± 5°C Initial limits | | Units | Fig |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| X20 | 5R (Cin) | 34 | Open | 2.0 | 18.0 | pF | 29 |
| X21 | 6D (Cin) | 18 | | | | | |
| X22 | 6S (Cin) | 38 | | | | | |
| X23 | 6C (Cin) | 39 | | | | | |
| X24 | 6R (Cin) | 37 | Open | 2.0 | 18.0 | pF | 29 |



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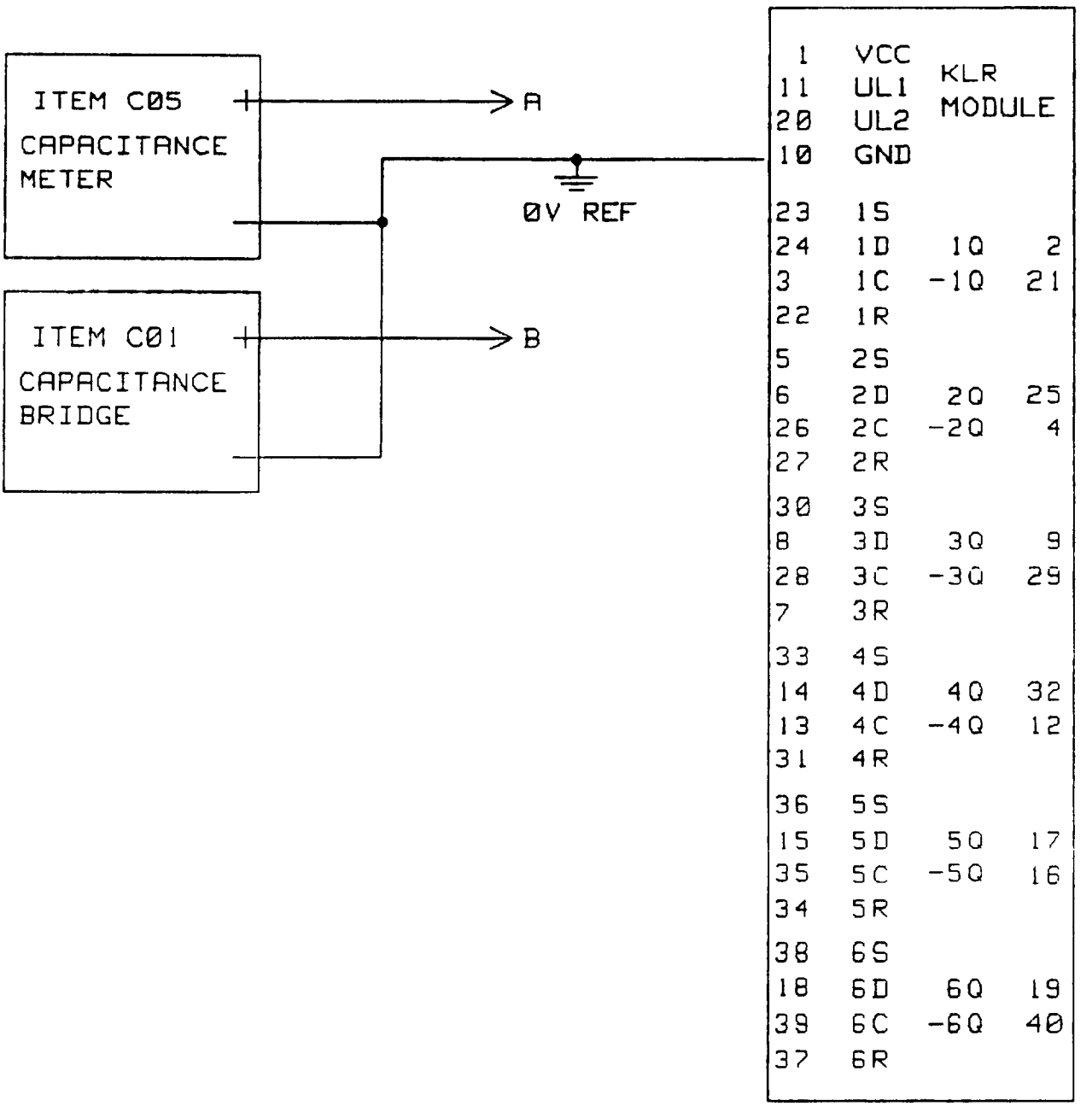



FIGURE 29. Power supply and input capacitance test circuit for table XXI.



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NOTES:

1. Null out the effects of the test leads prior to connecting to the module.

2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

3. Item C01 setup:

Frequency = 1.0 KHz.
Vpk = 0.3 V maximum.



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TABLE XXII. Functional test.

| Test cond | Parameter | Input pin connections (2 3 4 2 3 2 5 6 2 6 7 2 0 8 8 2 7 3 4 1 3 3 6 1 5 5 4 8 8 9 7 3) | Fig |
|---|---|---|---|
| Y1 | Functional | B B B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B | 30 |
| Y2 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y3 | | A A A B B B A A A A A A A A A A A A A A A A A A A A A A A A A A | |
| Y4 | | B B B A A A B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y5 | | B A B B A B A B A B A B A B A B A B A B A B A B A B A B A B A B | |
| Y6 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y7 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y8 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y9 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y10 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y11 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y12 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y13 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y14 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y15 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y16 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y17 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y18 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y19 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y20 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y21 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y22 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y23 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y24 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y25 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y26 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y27 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y28 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y29 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y30 | | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | |
| Y31 | | B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A | |
| Y32 | Functional | B B A A B B B B B B B B B B B B B B B B B B B B B B B B B B B B | 30 |



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TABLE XXII. Functional test - Continued.

| Test cond | Parameter | Outputs C to pin 2 | 21 | 25 | 4 | 9 | 29 | 32 | 12 | 17 | 16 | 19 | 40 | Fig |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Y1 | Functional | H | H | L | H | H | L | H | H | H | H | L | H | 30 |
| Y2 | | L | \| | L | H | \| | L | H | \| | L | \| | L | H | \| |
| Y3 | | L | H | H | L | \| | H | L | \| | L | H | H | L | \| |
| Y4 | | H | L | \| | L | H | \| | L | H | H | L | \| | \| | \| |
| Y5 | | \| | L | \| | H | L | \| | H | L | \| | \| | \| | \| | \| |
| Y6 | | \| | H | H | \| | L | H | \| | \| | \| | \| | H | L | \| |
| Y7 | | H | \| | L | \| | H | L | \| | \| | H | L | L | H | \| |
| Y8 | | L | \| | L | H | \| | \| | H | L | L | H | H | L | \| |
| Y9 | | L | H | H | L | \| | \| | L | H | H | L | \| | H | \| |
| Y10 | | H | L | \| | \| | H | L | H | L | \| | H | \| | \| | \| |
| Y11 | | \| | \| | \| | \| | L | H | \| | H | \| | \| | H | \| | \| |
| Y12 | | \| | \| | H | L | H | L | \| | \| | H | \| | L | \| | \| |
| Y13 | | H | L | L | H | \| | H | H | \| | L | \| | \| | \| | \| |
| Y14 | | L | H | H | L | \| | \| | L | \| | \| | \| | \| | \| | \| |
| Y15 | | H | L | \| | H | H | \| | \| | \| | \| | \| | L | H | \| |
| Y16 | | \| | H | \| | \| | L | \| | \| | H | L | H | H | L | \| |
| Y17 | | \| | \| | H | \| | \| | \| | L | H | H | L | \| | \| | \| |
| Y18 | | H | \| | L | \| | \| | \| | H | L | \| | \| | H | L | \| |
| Y19 | | L | \| | \| | \| | L | H | \| | \| | H | L | L | H | \| |
| Y20 | | \| | \| | \| | \| | H | L | H | L | L | H | \| | \| | \| |
| Y21 | | \| | \| | L | H | \| | \| | L | H | \| | \| | L | H | \| |
| Y22 | | L | H | H | L | H | L | \| | \| | L | H | H | L | \| |
| Y23 | | H | L | \| | \| | L | H | L | H | H | L | \| | \| | \| |
| Y24 | | \| | \| | H | L | \| | \| | H | L | \| | \| | \| | \| | \| |
| Y25 | | H | L | L | H | L | H | \| | \| | \| | \| | \| | \| | L | \| |
| Y26 | | L | H | \| | \| | H | L | \| | \| | \| | L | H | H | \| |
| Y27 | | \| | \| | L | H | \| | \| | \| | L | H | H | L | \| | \| |
| Y28 | | L | H | H | L | \| | \| | H | H | L | \| | L | H | \| |
| Y29 | | H | L | \| | \| | \| | L | L | \| | L | H | H | L | \| |
| Y30 | | \| | \| | \| | \| | H | H | L | H | H | L | \| | L | \| |
| Y31 | | \| | \| | \| | L | L | \| | H | L | \| | L | \| | H | \| |
| Y32 | Functional | H | L | H | H | L | H | H | L | H | H | H | H | 30 |

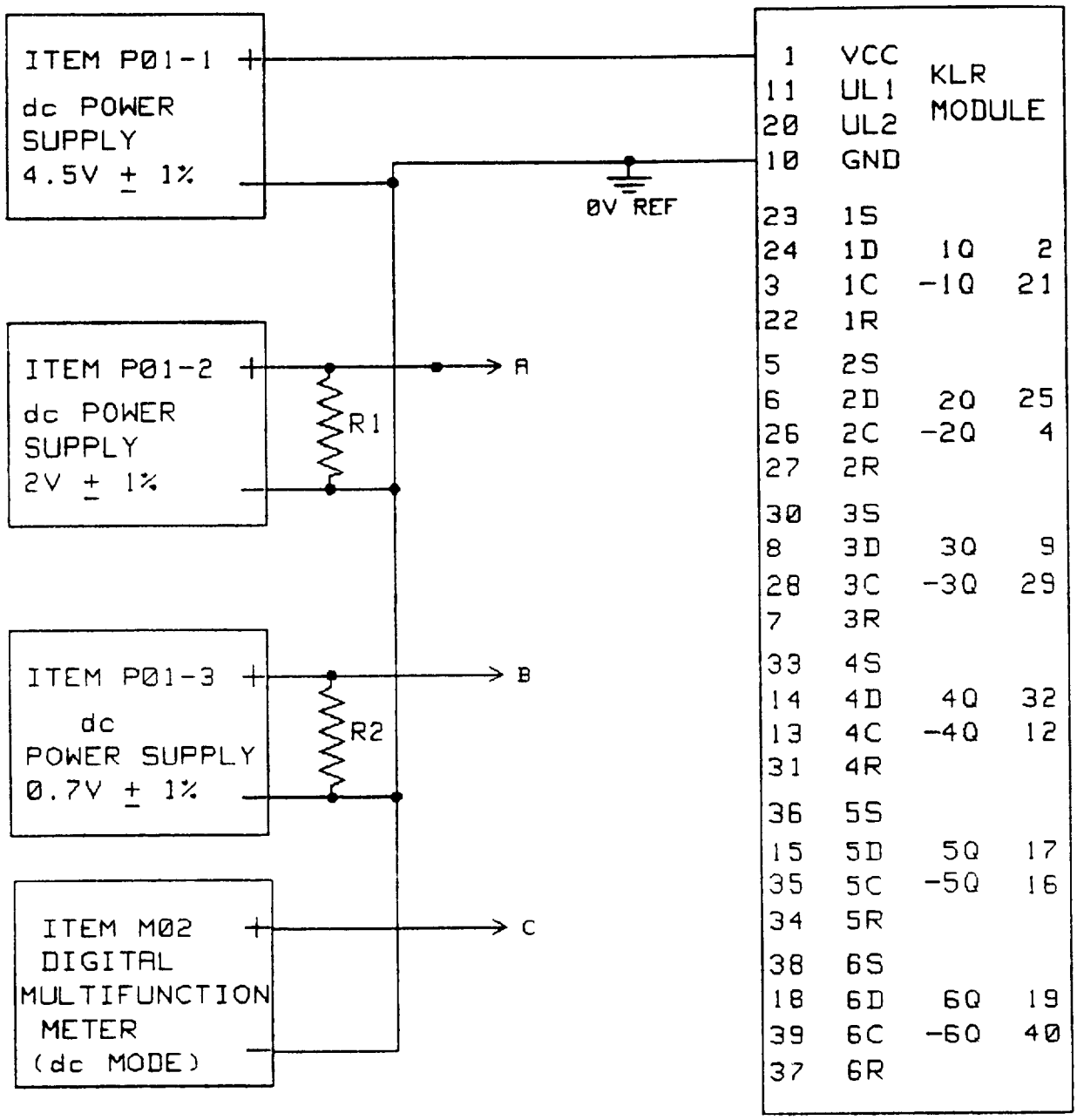
NOTES:

1. H = High level ≥ 2.5 V.

2. L = Low level ≤ 0.4 V.



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NOTE:

1. R1 = R2 = 51 Ohms ± 5 percent, 1/4 W.

FIGURE 30. Functional test circuit for table XXII.



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APPENDIX

MODULE FUNCTION

10. SCOPE

10.1 Scope. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. Module function. This is an edge triggered D type flip-flop with independent set and reset inputs with complementary Q and $\overline{Q}$ outputs. Input information on the data input is transferred to the output on the positive edge of the clock pulse. Clock triggering occurs at a voltage level during the rise transition of the clock pulse. After the clock input threshhold voltage has been passed, the data input is locked out. The set and reset inputs are independent of the clock. A low level input to the set line sets Q to a high level, and a low level input to the reset line sets $\overline{Q}$ to a high level. Simultaneous application of a low level to the set and reset inputs will force both outputs to a high level.



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## APPENDIX

30.1 Input to output relationship. Each D type flip-flop has the input to output relationship of figure 1 and table I for positive logic.

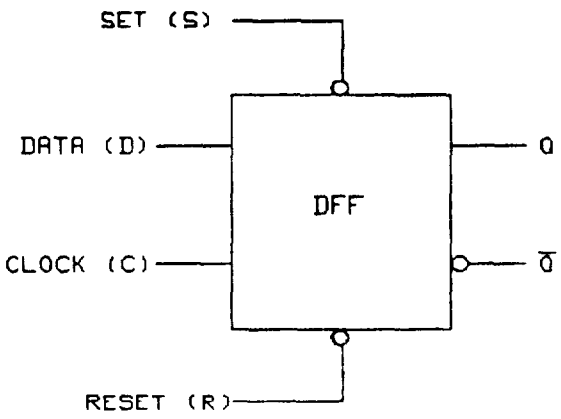



FIGURE 1. Logic diagram of functional circuits

TABLE I. Input to output relationship. 1/ - 5/

| INPUT $t_n$ | | | | OUTPUT $t_{n+1}$ | | |
|---|---|---|---|---|---|---|
| S | R | Clock | D | Q | $\overline{Q}$ | Operation |
| H | H | ↑ | L | L | H | Data entry |
| H | H | ↑ | H | H | L | Data entry |
| H | L | X | X | L | H | Reset |
| L | H | X | X | H | L | Set |
| L | L | X | X | H | H | * |
| H | H | L | X | $Q_0$ | $\overline{Q_0}$ | Hold |

1/ X - Don't care.
2/ $t_n$ - Time period before positive clock transition
3/ $t_{n+1}$ - Time period after positive clock transition
4/ ↑ - transition from low to high level
5/ $Q_0$ = the level of Q before the indicated steady state input conditions were established

*This configuration is nonstable, that is, it will not persist when set and reset inputs return to their inactive (high) level.



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| 1 DOCUMENT NUMBER | 2 DOCUMENT TITLE |
|---|---|
| MIL-M-28787/359(SH) | Modules, Digital, Standard Electronic, Flip/Flop, D Type Module, Key Code KLR |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4 TYPE OF ORGANIZATION *(Mark one)* |
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| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)* _____ |
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5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

| 7a NAME OF SUBMITTER *(Last, First, MI)* – Optional | b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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